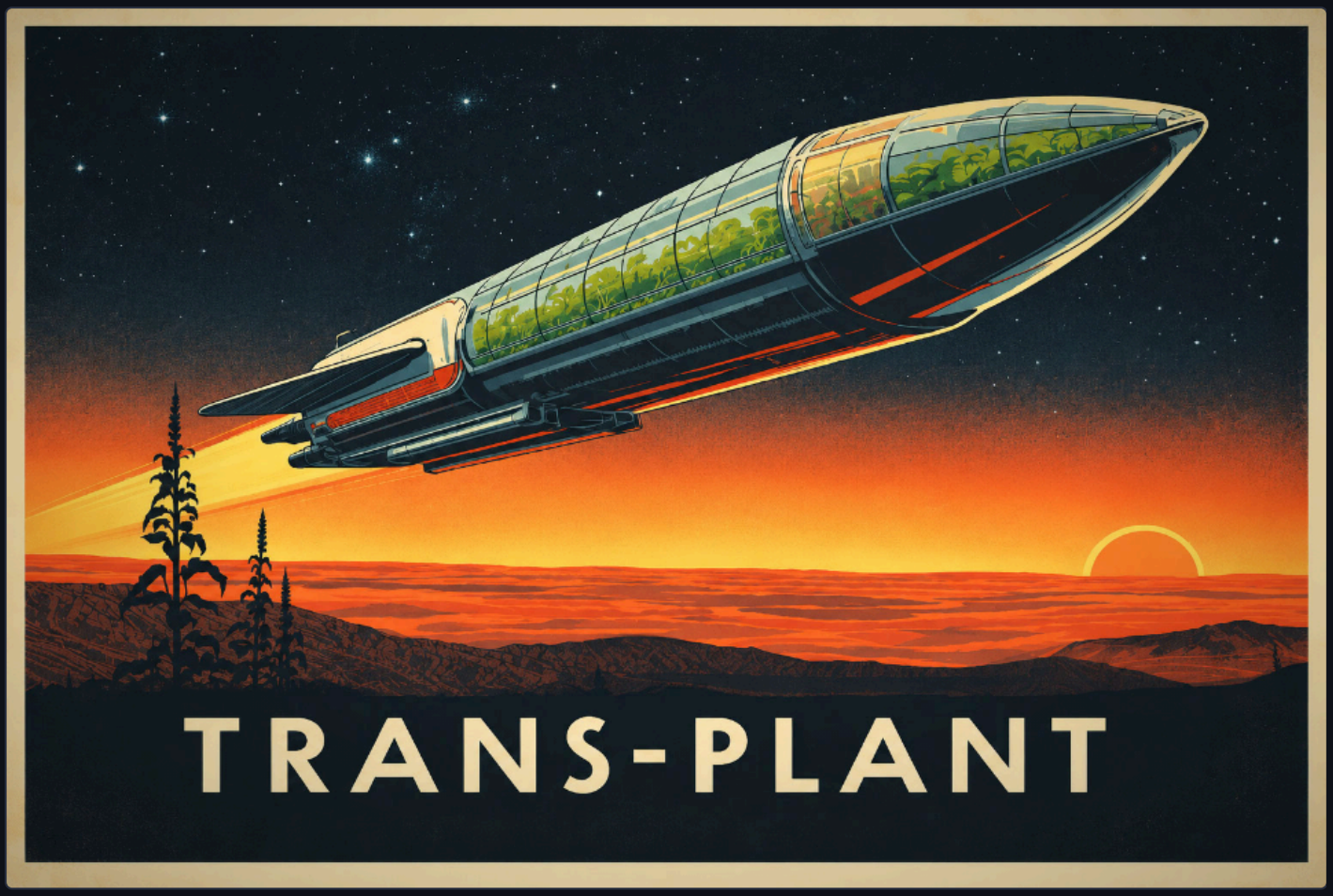
TRANS-PLANT

The Manual - how to play, and how it all works



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PART 1 · HOW TO PLAY

Everything a captain needs to know, with nothing spoiled that the trail should surprise you with.

The idea

Earth is dying. You have a ship, a crop, and a plan. The plan is not good.

You are the captain of a colony ship carrying one living crop to a new world. Your job is to get that plant across a freshly generated star map and into new soil, alive. The crop *is* the mission: you can arrive with a full crew, a shiny hull and a pile of gold, but if the plant in the growbay is dead, the colony has nothing to grow.

Between you and the destination are stations, planets, asteroid fields, nebulae, derelict ships, anomalies and a great deal of nothing. Every stop brings an event and a choice. Every choice costs something: days, supplies, morale, hull, gold. Your supplies drain as you fly, your crew gets tired and hurt, and the colony would like you to arrive before the deadline.

There are three trails, and you pick one at the start of every run:

Trail	Length	What it is like
Lunar	About three months, around a dozen stops, one fork	The gentle one. Established corridors and plenty of stations. A good first run.
Mars	About six months, around nineteen stops, two forks	You leave short on water and food, and the trail makes you work for every drop.
Interstellar	About ten months, around two dozen stops, three forks	The long haul to Far Garden. Stranger places, long empty stretches, and the hardest trouble saved for the end.

A run is a roguelike voyage: one life, no going back along the route, and a new map, new events and a new ending each time.



A run at a glance. Most of your time is spent in the loop between the hub, the next leg and the next event. A run can also end early, if something vital runs out.

Your first run

This walkthrough follows a real Lunar run from the title screen to the colony. Your run will meet different events, but every run follows the same shape.

1 · The title screen

The menu lives on the terminal of the ship's observation deck. Press any key (or click) to skip the opening lines. **NEW GAME** starts the setup wizard. **CONTINUE** picks up your autosaved voyage, if you have one. **RECORDS** lights up once you have finished a run and shows your past voyages and best scores. A button's lamp is lit when it can do something, and the **SHIP STATUS** screen below says what is aboard, or what the button you point at does. NEW GAME and CONTINUE play a short departure first; any key or click skips it. The deck's windows show one of four views, drawn fresh each time you load the game: Earth with its station, Mars, a ringed planet or a nebula.



The title menu, with Earth in the window. CONTINUE and RECORDS stay dark until there is a saved voyage or a finished run.

2 · The setup wizard

Six quick steps, shown as pips across the top. Pick a card on the right and its flavor line appears on the left. **NEXT** (or `Enter`) moves on, **BACK** goes back, and on the first step **MENU** returns to the title. Each card states what it does in its green stat line, and those numbers are the game's real ones.



Trail: how long, how many forks, how hard.



AI: your narrator, with one passive perk.



Crew: up to two specialists, or none.

- **Trail.** Lunar, Mars or Interstellar (see The idea). Longer trails mean more stops, more forks, tighter supplies and harder events.
- **AI.** The ship computer narrates the whole voyage, comments on everything you do, and gives one passive perk. See The five AIs.
- **Captain.** That is you. Your background is what you studied, and it gives you a half-strength version of one specialist's skill, sometimes with a starting item or extra gold. See Captains.
- **Crew.** Pick up to two of seven specialists. Each one has an always-on passive skill, an active skill you trigger yourself, and choices in events that only they can unlock. Picking fewer than two asks you to confirm; picking none means flying solo. See Crew roles.
- **Crop.** What you are keeping alive: wheat, tomatoes, sweet potatoes, soybeans or zinnias. They differ in how fast they grow, how thirsty they are, what they resist and what a harvest gives. See Crops.
- **Ship name.** The ship arrives with a name from the colony registry. **RE-ROLL** draws another, or type your own. Then **LAUNCH**.

> **A good first run:** the Lunar trail, any AI that amuses you, and a botanist or an engineer in the crew. Sweet potatoes or wheat forgive mistakes. Tomatoes and zinnias are for when you want drama.

3 · Departure, at the station

After LAUNCH the voyage starts on day 0 at Earth Orbital Station, and something happens before you have even left: the **departure event**. Events are the heart of the game. A scene and a story panel fill the screen, the text types itself out, and once it has landed your choices appear on the right. Click anywhere on the story to make the text finish at once.



The departure in this run. The third choice is locked (it needs a diplomat in the crew), and its two filled pips show the effort it would take.

As you leave, your AI also notes in the log whose space this route crosses: every voyage passes through the territory of one alien species (see Species space and stations).

The five dots at the top right are a compact gauge you see during events: fuel, food, oxygen, hull and the crew's average morale. The line at the top left is your AI's running commentary. We will make this choice in step 6.

4 · The cruise hub

When the departure is over, you land on the **cruise hub**, the screen you come back to between every stop. The ship is still parked at the station, so the hub opens *stopped*: the **Stop menu** sits in the right-hand column, and the AI says its piece in a speech bubble over the ship.



Day 1, stopped at the station. The bottom row is your ship at a glance: crew, crop, supplies, and the STOP/RESUME button.

Everything on this screen is explained in The screens, one by one. For now, the key parts are:

- the **header**: ship name, a news ticker, your gold, the cargo hold and map buttons, and the day counter with the days left before the deadline;
- the **standing orders** strip under the ship (engines, rations, water) which opens ORDERS;
- the **crew strip**, the **crop card** and the five **supply bars** along the bottom;
- the big **STOP / RESUME** button in the bottom right corner.

Your AI also gives first-run tips like the one in the bubble, one per stop, the first time each topic comes up. You can switch them off in the pause menu's SETTINGS.

5 · Stopping: the Stop menu

Every time you arrive somewhere and finish its event, the ship is stopped there and the Stop menu opens. Its jobs depend on the kind of place: at a habitable planet you can GATHER food, SCAN ahead, TRADE if a trader is in range, REST the crew or run the FABRICATOR.



Stopped at a planet with a trader in range. Each job shows its effort pips and how long it takes; hover a job for its morale cost.

Every job costs time, and most cost a little morale (the effort tax, see Events and choices). You can do as many jobs as you like at a stop, one after another. When you are done, press **RESUME** to fly on. Doing nothing is also a choice: the deadline is always ticking.

6 · An event, and making a choice

Back to that departure. Hover over a choice to see what it really costs: the days it takes and any supplies it spends. Here, signing up a breakfast-cereal sponsor takes a day and costs a little morale.



Hovering a choice shows its time and resource costs.



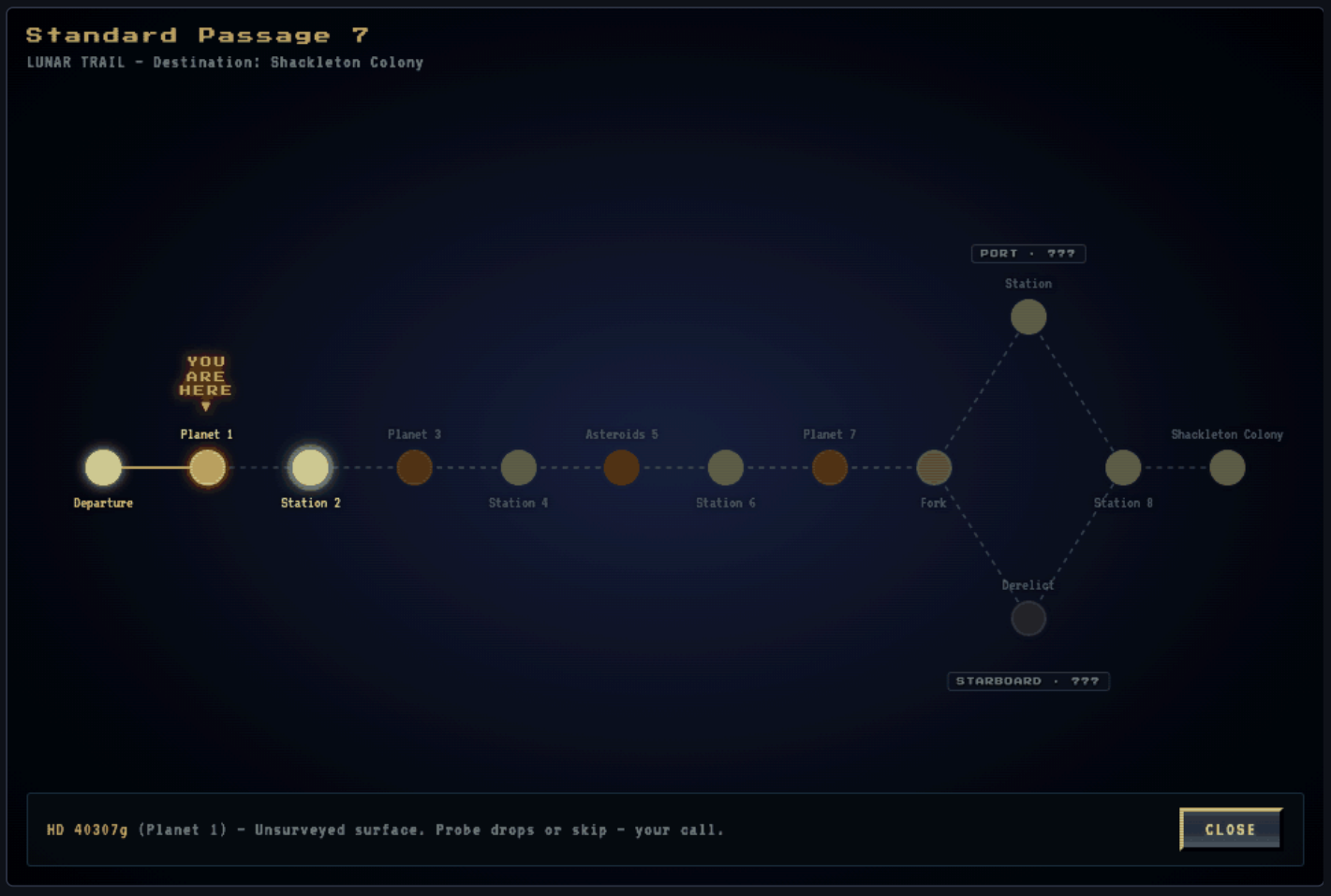
The outcome. CONTINUE takes you back to the hub.

Click a choice (or press its number key, `1` to `9`) and the outcome plays. What it did to the ship is summed up in the AI log and the Ship's Log: in this run, "+12 GOLD, −2 MORALE, 1 DAY". Morale tops out at 100; when a lift has nowhere to go, the AI says so in the log. Some choices lead to a second set of choices before the outcome; some start a hands-on minigame or a ship-to-ship fight.

7 · Traveling on, and forks

Press **RESUME**. The ship flies the leg to the next stop: the stars streak past at the speed your ENGINES order sets, the days of the leg pass, and fuel, food, water and oxygen are drawn down. You arrive, the stop's event plays, and you are stopped again. That is the rhythm of the whole voyage.

The route is a line of stops with a few **forks**. At a fork, an event asks you to pick a lane, and you cannot come back for the other one. The **map** button in the header shows the whole route; lanes you have not chosen yet are marked with question marks.



The route map early in a Lunar run. Click any stop for a one-line description.

> A good SCAN just before a fork reports what waits at the first stop down each lane.

8 · A station visit

Stations are where you shop, heal, repair and recharge. Arriving at one opens the **approach** on the bridge with the station in the window. Sometimes the dockmaster calls first; otherwise **HAIL** opens comms. Then **DOCK**. Some stations charge a small parking fee, and DOCK says so.

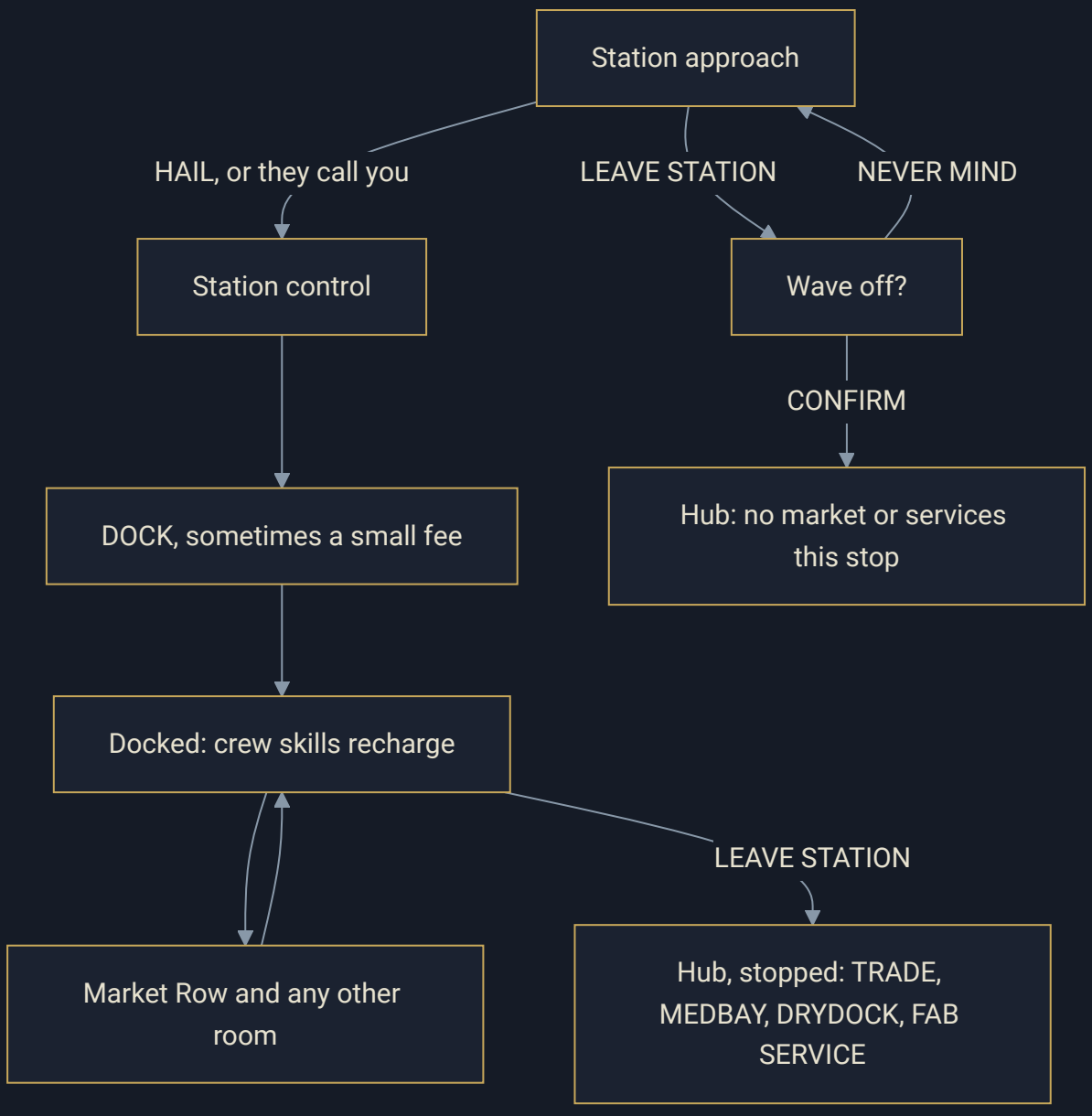


The approach. LEAVE STATION here means waving off: no berth this time.



Docked. MARKET walks to Market Row; LEAVE STATION goes back to the bridge.

Once docked, your crew's active skills recharge. Market Row is always there; sometimes there is also someone in another room (a cantina, a clinic, an office) with something to say. Each room plays once per visit. When you **LEAVE STATION** you are back on the hub, stopped at the station, and the Stop menu now includes the **station services** you can pay for: MEDBAY, DRYDOCK and FAB SERVICE (each only shown when you need it), plus TRADE to go back to the market. Some stations further out also keep a **BACK ROOM** (see The back room).



How a station stop flows. Waving off costs no time, but closes the station's market and services for this stop.

9 · Arriving

Reach the last stop and the voyage is over. The arrival screen shows the colony with your crop, under a title that tells you how well it went. **VIEW REPORT** opens the report card.



The arrival. The scene shows your destination and your crop as it arrived.



The report card for this run: everyone home, fourteen days early, and a new best for the trail.

The report card shows:

- the **tier title**, an arrival line about your crop at its new home, and a closing quote;
- the **Run Report**: how many of the people who launched are still aboard, how full your fuel, food, water, hull and morale were, the day you arrived against the deadline (days early, on time, or days late), and your **score**, with **NEW BEST** when it beats your record for that destination;
- score extras, when you earned them (soil surveys, a crop bonus);
- **READ THE COLONY REPORT**, which unfolds the full ending text and the **Ship's Record**, the most notable things you did on the way;
- **NEW RUN**, and back to the title.

A run that ends early (see How a run can end) gets its own screen with an epitaph and the Ship's Record instead.

The screens, one by one

The cruise hub and its HUD



The hub while flying. With the ship moving, the right-hand column is the AI log; when you stop, the Stop menu takes its place.

Part	What it shows and does
Ship name (top left)	The name you gave her. The AI will use it, a lot.
Ticker (top middle)	What is happening right now: `COMING UP ON …` while flying, `STOPPED` at a stop, `TRADER IN RANGE` when one is nearby, and the news from a good SCAN.
Gold, cargo, map	Your purse, then the crate button for the cargo hold and the chart button for the route map.
Day counter (top right)	`DAY 1 · 99 LEFT`: today, and the days left before the deadline. Past the deadline it counts the days late instead.
Ship view	Your ship, the starfield (faster engines, faster stars), the place you are stopped at, and alert labels when something needs attention.
Standing orders strip	ENGINES, RATIONS and the WATER split between plant and crew. Click it to open ORDERS.
AI log (right column)	Everything that happens, in your AI's voice, under day dividers. Lines with a link icon carry a real-science reference.
Crew strip	Three same-size slots: the captain and up to two crew. HP (health) and MO (morale) bars; hover a bar for the exact number and, for morale, why it moved. The captain has no morale bar. Click a card for the crew dossier. An empty slot reads "vacant".
Passenger line (bottom left of the ship view)	`PASSENGER` or `PASSENGERS` and their names: anyone riding along without a crew place. Hover it for where each one gets off. It is only there while someone is riding. See People who come aboard.
Crop card	Your crop: its stage, its day against its days to maturity, growth and health bars, and an alert light. Problems show in amber (`THIRSTY`, `SICK`, `PESTS: TREAT IT`). When it is ready a HARVEST button appears. Click the card for the growbay.
Supply bars	`FOOD` `H2O` `O2` on the left, `FUEL` `HULL` on the right. Hover for the exact level.
STOP / RESUME	STOP while flying opens the Stop menu; RESUME while stopped flies on to the next stop.

The AI's speech bubble

While you are stopped the Stop menu covers the log column, so your AI talks to you in a **speech bubble** over the ship instead: its portrait and its newest three lines for this stop (tips, comments, warnings, memories of earlier choices). Click the bubble or its **×** to dismiss it; RESUME clears it too. Every line is also kept in the log and the Ship's Log, so nothing is lost. You can see the bubble in the walkthrough's hub screenshot. Hovering a Stop-menu job fades the bubble while the job's box would cover it.

The Stop menu

The Stop menu lists the jobs this place allows. Its header names the kind of place (DEEP SPACE, ASTEROID FIELD, NEBULA, STATION APPROACH...) with a line of mood, and a **LOG** button that opens the Ship's Log. Each job row shows its effort pips, how long it takes, and a one-line description. Hover for the details, including the morale the job will cost each crew member. A locked job says what it needs. Jobs with nothing to work on (REPAIR with a full hull, MEDBAY with nobody hurt) are not listed at all.



At a station with a hurt crew member, a dented hull and a worn fabricator: the paid station services appear.



REST and WAIT FOR TRADER ask how many days, one to seven.

Job	Where	Time	What it is for
MINE	Asteroid fields, nebulae, derelicts	1 day per run	The mining minigame: crack rocks for materials. In an asteroid field some ice melts into water; in a nebula you scoop fuel. After a run you can MINE AGAIN for another day.
SALVAGE	Derelicts	1 day	The salvage minigame: tractor-beam loot into your hold. A derelict still in one piece is boarded (fuel cells, ration cans, medkits, parts); a broken one is swept from outside.
FORAGE (GATHER, HUNT, SCAVENGE)	Planets, asteroid fields, derelicts, anomalies	1 day	The forage minigame: find food. The label changes with the place. On an *alien* world it is FORAGE in a red risky row: someone aboard may be poisoned.
TRADE	Stations; elsewhere only when a trader is in range	0–2 days	Opens the market. Browsing is free; leaving after you bought or sold costs a day of haggling.
WAIT FOR TRADER	Most places, when no trader is in range	1–7 days	Park and hope a trader turns up. Each day might bring one, and other things can hear you waiting too. STOP WAITING calls it off (days spent stay spent).
SCAN	Deep space, planets, nebulae, anomalies	1 day	The scan minigame. A decent result reports what waits at the next stop (and down every lane of a fork); a clean lock looks two stops ahead and shows what each choice there will do.
REPAIR	Anywhere, when the hull is damaged	1 day	The engine-room repair minigame. Costs 2 metal and 1 scrap; patches some hull (more with an engineer).
REST	Anywhere, with crew aboard	1–7 days	Lifts every crew member's morale each day. Locked when everyone is already in good spirits. WAKE UP ends it early.
CONDENSE	Nebulae, with an Atmospheric Condenser	1 day	Wring water out of the gas.

MEDBAY	Stations, when someone is hurt	No days	Treats everyone aboard who is hurt, for gold.
DRYDOCK	Stations, when the hull is damaged	No days	Station crews patch the hull, for gold (less when there is less to fix).
FAB SERVICE	Stations, when the fabricator is worn	No days	Resets the fabricator's wear to zero (and fixes it if broken), for gold.
BACK ROOM	Some stations, past the first stretch of the trip	No days	Opens the black market: see The back room. Not every station has one, and the map does not say which do.
FABRICATOR	Anywhere	Per recipe	Opens the fabricator.

> **Starting a job and backing out.** Minigames open on a briefing. Closing the briefing before you press BEGIN is free, and the morale tax comes back. Quitting *after* BEGIN keeps nothing you found, and the day is still spent.

Events and choices

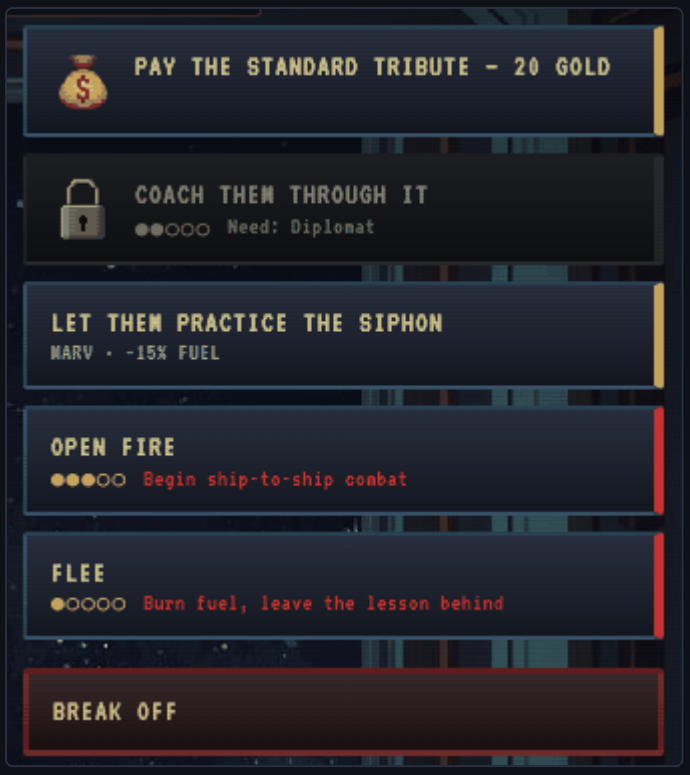
Every stop has one event. Some are quick decisions, some have several rounds of choices, some put you face to face with someone on a screen or in a corridor, and some turn into minigames or fights. The choices column is where the decisions are made, and it tells you a lot if you know how to read it.



Hands-on jobs: pips, the days and the clean odds.



A hands-on job's tooltip: who's helping, what it costs, and how it goes.



Gates, a locked choice, an AI preview, risky choices and the leave row.

- **The sub-label** is the small line under a choice. It is always one line, and the full story is in the tooltip. It may say what the choice needs, preview what it does, or describe the job.
- **Effort pips** ●●○○○ show how much work a choice is, from one to five. **Every pip costs morale**: each crew member pays the effort tax when you pick it, and more pips cost more morale, climbing faster than the pips do. The cost is in the tooltip, on Stop-menu jobs and event choices alike. With no crew aboard there is nobody to tax.
- **Hands-on jobs** show `2 days · 90% clean`: the time the job takes and the chance it goes cleanly. Your own hands, every crew member, the right specialist and the right tool all count toward the odds; the tooltip's HELPING line names who and what is on the job, and it says what missing help would change ("A portable drill would make it 90%."). A one-use tool (a repair kit, say) is spent when it counts. Short-handed, the job still gets done, just worse: the tooltip spells out exactly what "worse" costs. Even a well-equipped crew is never certain.
- **Locked choices** show a padlock and `Need: …`: a crew role, an item, an amount of gold, or several at once. A crew member who is too badly hurt cannot help, and the lock says so. Some choices stay hidden entirely until something earlier in the voyage opens them.
- **Gate icons.** An unlocked choice that uses something you have shows its icon: the crew member's portrait, the item, the gold.
- **Previews.** Some choices show what they will do before you pick them. Your AI might (MARV flags the most expensive option), a clean SCAN does, some items do, and an Academic captain gets one research preview per station leg.
- **Risky choices** have red sub-labels: they can go badly.
- **Tooltips.** Hover any choice for the full picture, to the left of the choice with an arrow pointing at it. The top box says what the choice is. The WHAT IT TAKES box below shows the days as blocks (solid for certain, hollow for maybe), the effort pips, who's helping, every cost as a red chip, and a red ▲ RISKY tag when it can go badly. A choice left to chance, or a hands-on job, adds one bar split by the odds with each result named, its chance and its biggest effects. Stop-menu jobs and combat actions use the same box, with a green GAINS row where the job says what it gives; a day in a job's day picker has its own. While a box covers the AI's speech bubble on the cruise screen, the

bubble fades until you move off. If the days a choice takes would run out your food, water or air, the box says so in red under the days, and if a supply is already gone and the wait would end the run, it says that.

- **Leaving.** Many layered events have a red leave row at the bottom (LEAVE, NOT NOW, BACK TO DOCK…). It ends the scene with no outcome, unless its own sub-label or tooltip says otherwise: some leave rows are an exit with a price (backing off from a fight is never free). A few emergencies cannot be left, and their leave row is locked.

Many choices also take time. A quick decision is a day; real work can be several days; a few big undertakings take a couple of weeks. The tooltip always says.

ORDERS

Your standing orders apply every day until you change them. Click the orders strip under the ship (or a readout in it) to open ORDERS. Each row shows a projection that updates as you try options; **APPLY** keeps the change, **CANCEL** throws it away.



ORDERS. Every option comes with its own projection and an AI comment.

- **ENGINES** sets the pace. **LIGHT SPEED**: crawl, burn little, the crew rests. **RIDICULOUS**: the default. **LUDICROUS**: faster, burns harder, the crew tires. **PLAID**: fastest, bleeds fuel and crew, and running at Plaid leg after leg heats the drive. The row projects your **ARRIVAL** day against the deadline, the day the fuel would run out, and the morale trend.
- **RATIONS**: **VENDING MACHINE** (half rations: food lasts, morale sinks), **STANDARD** (morale holds), **PIG OUT** (extra: morale lifts, food empties fast). The row shows food per day and the day the food runs out.

- **WATER ALLOCATION** splits the ship's water between the plant and the crew. Drag the slider. The marker shows what the crop needs; below it the crop is thirsty and slows, too far above it risks root rot, and starving the crew's side hurts morale and eventually health. The row projects when the plant matures and how the crew is doing.

> The ARRIVAL projection is the same one the game uses to decide you are behind schedule. If it shows a ⚠ and a day past the deadline, you are on course to arrive late.

The cargo hold

The crate button in the header opens the hold. The top shows your fuel, food and water tanks with their level, daily use and the day each would run empty. The food tank's daily use counts everyone aboard, passengers included. Below are three tabs: **GEAR** (instruments, consumables, trade goods), **MATERIALS** and **UPGRADES**, with a SORT menu. The header counts slots used against capacity.



Click any tile for its detail: what it is, what it costs in slots, what a trader would pay, and which situations it helps in.

From the detail panel you can **USE** a consumable (a fuel cell, a ration brick, a crop treatment…), **SELL** it to the market while you are docked (no day spent), or **JETTISON** it to free a slot. Upgrades work just by being aboard. A very few things refuse to be jettisoned, and the hold tells you so when you try.

The crew dossier

Click a crew card for their dossier: portrait, health and morale, a short bio, any status effects (wounded, morale crisis, deserting, going hungry or thirsty…), their **PASSIVE** skill (always on) and their **ACTIVE** skill with its uses and a **USE** button. Active skills recharge when you dock at a station. Arrows flip between crew members, and the captain has a shorter dossier of their own.



The dossier. DISMISS CREW lets someone go for good.

The growbay

Click the crop card to visit your crop. The growbay shows growth and health bars, the plant itself at its current stage, the stage name, the day the harvest will be ready, who is tending the plant, and every growth modifier in play. Under MONITORED are the threats that are not happening right now.



The growbay. The water slider here is the same one as in ORDERS.



INSPECT: the full read on your crop, with every active threat's cause and fix.

- **Water.** Set the plant's share of the water. The plant's need is marked on the track.
- **Tender.** Who looks after the plant. A botanist tends by default and is by far the best at it; anyone else helps a little. Click the tender card to change. Someone who joined on the way can tend too; for now they stand in the bay as a still portrait.
- **TREAT** clears a pest outbreak. It takes a day of work unless a botanist handles it; the button's tooltip says which.
- **INSPECT** opens the full read: vital signs, growth rate, species notes, the four stages and what can go wrong in each, and every active threat with its cause and fix.
- **HARVEST** unlocks when growth reaches 100%. It gives food (zinnias give morale instead) and the plant starts a fresh cycle. Food that will not fit in the stores is pressed into ration bricks if the hold has room.

The fabricator

The fabricator turns materials into useful things. Open it from the Stop menu. Pick a recipe on the left to see what it makes, what it needs (with what you have), how many days it takes, how much wear it adds, the chance of a misfire and its morale cost. **CRAFT** starts the job.



The fabricator. Two recipes stay locked until you find the right upgrade.

- **Crafting takes real days**, one at a time, while the fabricator is open, and the ship's usual daily drain applies. Close it and the job is held until you come back.
- **CANCEL** before the first day finishes gives everything back; after that you get half the materials back.
- **Locked recipes.** Two recipes need a fabricator upgrade, and two more need a blueprint you may come across on the trail. A blueprint takes no cargo slot; carrying it unlocks its recipe.
- **Wear** builds with every print and makes misfires more likely. At full wear the fabricator breaks. An engineer reduces wear, and a station's FAB SERVICE resets it.

The market and Market Row

At a station, TRADE (or the MARKET room when docked) takes you to **Market Row**, where a vendor runs a stall; out on the trail, TRADE hails a passing trader ship. **BROWSE THE WARES** opens the market.



The market. Left column: what they sell. Right column: what you can sell them.

- **BUY** rows sell an item for gold. Station markets stock a wide catalog, including ship upgrades; a trader ship carries a smaller hold.
- **TRADE** rows are barter: they take one of your goods instead of gold.
- **SELL** pays gold for your items and materials. Trade goods each have their own value; a spectrometer aboard tells you exactly what it is before you sell.
- Discounts (CHIP, a Merchant captain) apply to what you buy, never to what you sell, here and in the back room.
- **STEP BACK** closes the market. Browsing is free; leaving after you bought or sold costs a day.

The back room

Not everything for sale is on Market Row. Some stations past the first stretch of the trip keep a **BACK ROOM**, and every voyage passes at least one. It shows up as a job in the Stop menu once you have docked: two effort pips and no days. A station you wave off keeps its back room shut along with its other services. A few people you meet in a bar, or out on the trail, may also offer to show you what they keep off the books. Either way you reach the same **black market**, with the hub dimmed behind it.

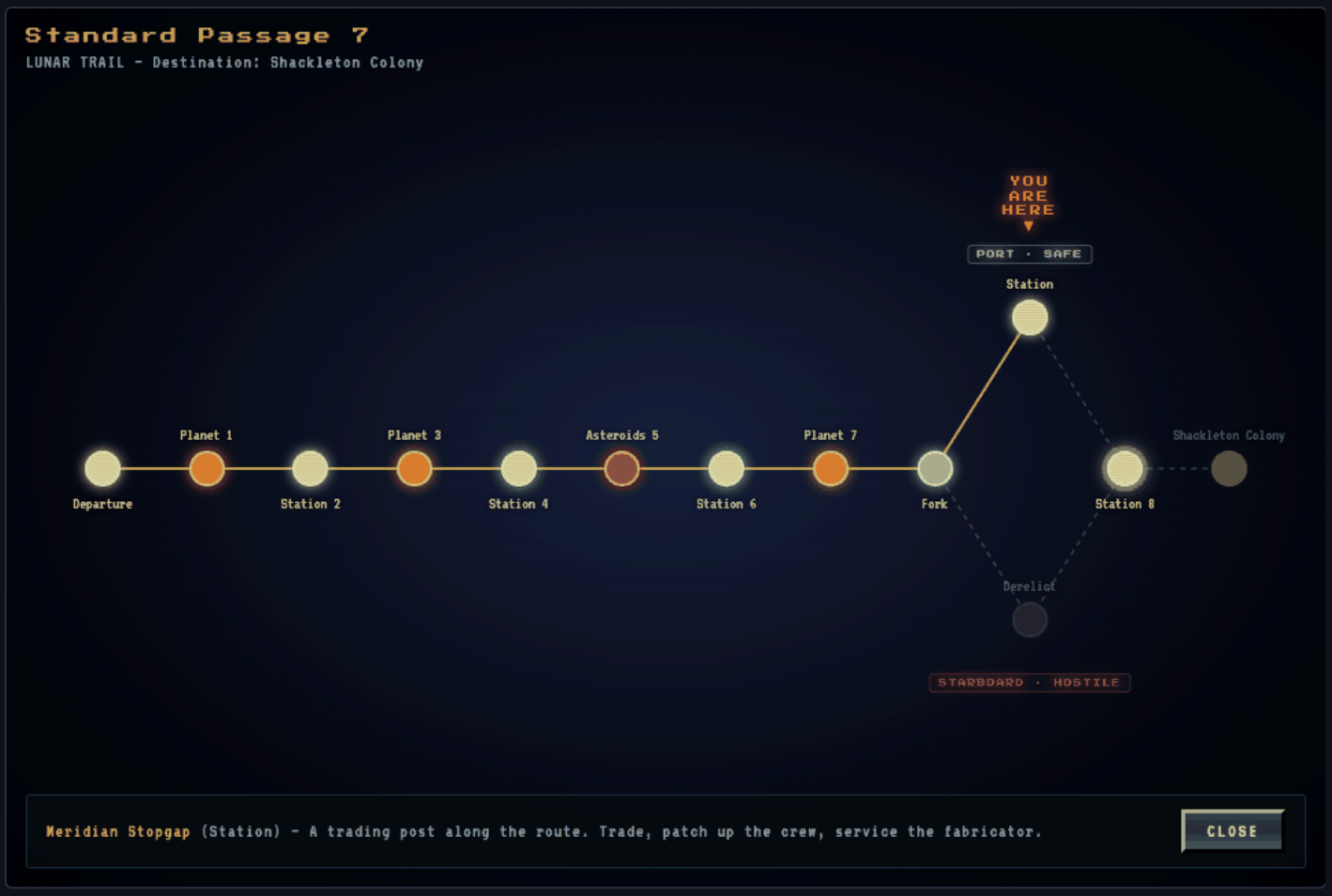


A back room at a reptile station. The left column is what they sell; the right is what they will fence.

- **Off-manifest gear.** A few ship upgrades no honest shop shelves. One you already carry is not offered again.
- **Gray-market supplies.** Each seller keeps two of five kinds: bigger refills of oxygen, fuel, hull, food or healing than the honest versions. They are cheap for what they promise, and sometimes they go wrong when you use them. The tooltip says what can go wrong, but not how often. They stack like other consumables, so you can buy several. Harm from one that goes wrong counts like any other harm: if it takes the hull or morale to zero, the voyage ends.
- **A buyer for odd cargo.** The back room buys trade goods and curiosities, and pays more for them than an honest trader. It has no interest in your materials or your gear.
- CHIP's and a Merchant captain's discounts apply here too. **LEAVE.** closes it.

The route map

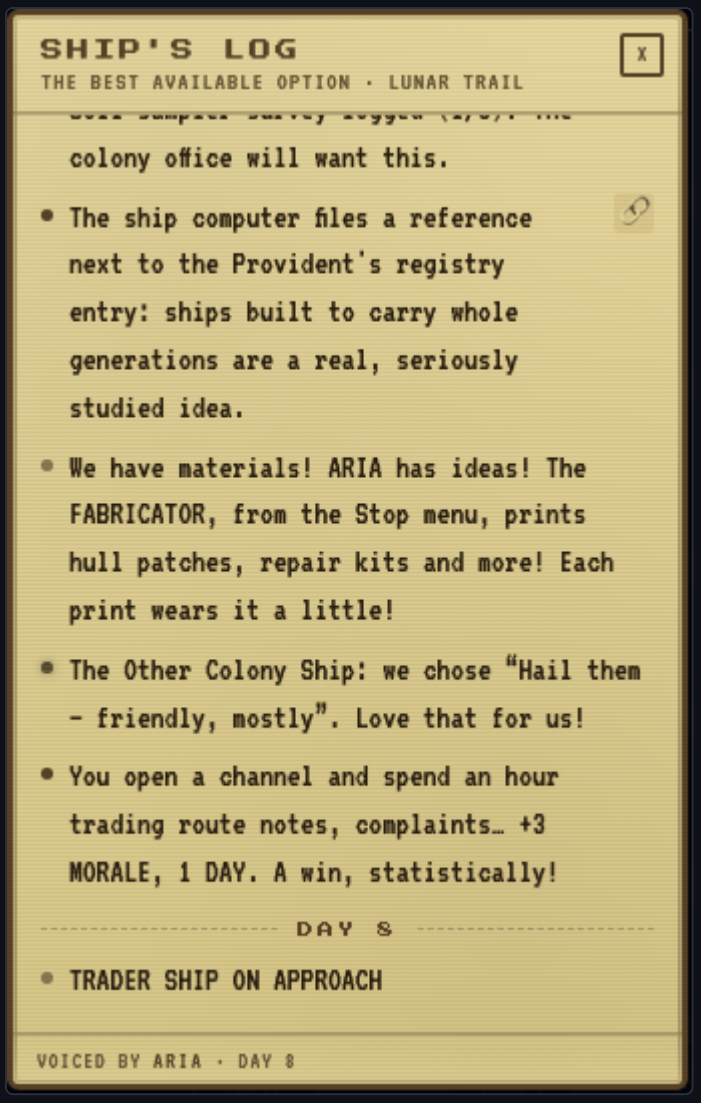
The map button in the header shows the whole route: every stop, colored by type, from Departure to your destination, with **YOU ARE HERE** over the ship. Stops you have passed are marked visited. Click a stop to read what it is. Forks split into named lanes; until you pick, the lanes beyond them read `???`. The map is for looking only: you choose lanes in the fork's event. A station that belongs to one species is labelled with the species' name instead of "Station" (Reptile 11, say); its shape still marks it as a station.



The same map after the fork. The lane not taken stays dimmed.

The Ship's Log

The **LOG** button in the Stop menu (or SHIP'S LOG in the pause terminal) opens the voyage's journal, written by your AI. It records every event you met and what you chose, what it really did to the ship, every arrival and what the leg cost, crop milestones, fights, and people joining or leaving. Entries with a link icon point to the real science behind a joke. Anyone lost on the voyage gets a memorial page.



The Ship's Log, voiced by whichever AI you chose.

The pause terminal

Press `Esc` any time for the **Bridge Terminal**. Time stops while it is open. It opens on a short ship status report: who is aboard (a captain with only passengers left is not reported as alone), the crop, the day and the size of the Ship's Log.



SETTINGS.



HELP: the controls in one place.

- **RESUME** closes the terminal.

- **SETTINGS**: cursor color and outline, **SHOW TIPS** (the AI's first-run tips), and **TEXT SPEED** for the typewriter text.
- **SAVE** and **LOAD**: three local save slots. You can save between encounters (not in the middle of an event or a job), and a loaded game returns to the hub. The game also **autosaves** as you play; CONTINUE on the title screen resumes it.
- **SHIP'S LOG** opens the journal.
- **HELP** lists the controls: click to act, hover to preview, drag sliders, `Esc` for this terminal or to close a window, `1`–`9` to pick a choice or a Stop-menu job, `Tab` and `Enter` to move between buttons and press one.
- **QUIT** returns to the title screen after a confirmation. Anything since the last save or autosave is lost.

> Your saves live in your browser, on this computer. Clearing the browser's site data clears them too.

Records

RECORDS on the title screen lists your **best arrival per destination** (score, tier, ship, crop, captain, day and crew; the crew reads like "crew 2/2", or "crew 2 (launched with 1)" when newcomers outnumber the losses) and your **last 10 voyages**, newest first: arrivals with their tier and score, and runs that ended early with how they ended. Only arrivals whose crop made it alive count as a best. **CLEAR RECORDS** wipes them, after asking.



Records after one finished Lunar run.

What everything means

The seven resources

Fuel, food, water, oxygen and hull are ship supplies, each out of 100 (two upgrades raise the fuel and hull caps). Morale belongs to each crew member. Gold is money. The supply bars turn to warning colors when a supply runs low, and your AI will mention it.

Resource	What runs it down	What brings it back	If it hits zero
FUEL	Every leg you fly. Faster engines burn much more. Some choices and escapes.	Fuel cells, MINE in a nebula, salvage, traders, events. A pilot, a pilot captain or a drive coil make each leg cheaper.	You are adrift and the run ends, unless something aboard can call for help.
FOOD	Every day, by your rations and the number of mouths (the engineer eats twice as much; passengers eat too). Some events.	Harvests, FORAGE, ration bricks, salvage, the market. A chef makes food last longer.	Everyone starves: health and morale fall each day, and too many days without food ends the run.
H2O	Every day, shared by plant and crew. Some events (on Mars, events ask a lot more of it).	Water canisters, ice from MINE in an asteroid field, CONDENSE, a water purifier at planets, events, traders.	Everyone goes thirsty and the crop dries out. Too many dry days ends the run.
O2	The crew breathes it, slowly. Some accidents vent it.	O2 canisters and events. A healthy crop makes some of your air, so the draw is lower while the plant thrives.	The fastest way to die: only a few days.
HULL	Accidents, impacts, storms, fights.	REPAIR (metal and scrap), DRYDOCK (gold), hull patches, some events. An engineer gets more out of every repair.	The ship breaks apart and the run ends.
MORALE	Effort pips, short rations, hard pace, bad news, wounds, losses, and things each specialist personally minds.	REST, good rations, good news, harvests (especially zinnias), a chef's cooking, ARIA.	A crew member at zero may desert; the whole crew at zero mutinies (see below).
GOLD	The market, station services, parking fees, tributes, bribes, and some choices.	Selling items and materials (salvage is the steady earner), event rewards, loot from fights.	Nothing, except that you cannot pay for things.

Every run starts with 50 gold (more for some captains), a repair kit, a soil sampler, and a little metal and scrap. Mars starts short on food and water; Interstellar starts a little short on both, with a crew already less cheerful about the trip.

Crew roles

Each specialist brings a passive skill (always on), an active skill (one use, recharged when you dock at a station), extra effort for jobs in their line, and choices in events that only they can unlock. Anyone aboard adds a pair of hands to hard work.

Crew member	Good at	Active skill
Dr. Osei, Botanist	Crop care. The crop drinks less, grows and yields more with Osei tending, and TREAT takes no day.	**Greenhouse Protocol**: restores crop health.
Kazuki, Engineer	Repairs: event repairs go further, REPAIR patches more, less fabricator wear, a hull patch in combat. Eats twice as much.	**Emergency Patch**: stops one hull-damage hit.
Dr. Vasquez, Medic	Keeping people alive: deadly outcomes more often become injuries, and the hurt heal faster.	**Treatment**: heals one injured person.
Lindqvist, Pilot	Fuel: every leg burns less.	**Push Engines**: skip the next stop entirely, at double fuel.
Reyes, Chef	Food: it lasts longer, and real meals lift morale a little every leg.	**Special Meal**: a morale boost, paid for in food.
Dr. Tanaka, Xenobiologist	The strange: extra, safer options at anomalies and around alien life; delighted by alien finds.	**Field Analysis**: learn what an unknown biological item really is before you commit.
Hargrove, Diplomat	The one specific situation: blunts the first enemy volley, and pirates give up sooner.	**Negotiation**: opens a peaceful way out of hostile encounters.

HP, wounds and morale

- **Health (HP)** runs from 100 down. Below 40 a person is **wounded**: their active skill goes offline, everything else they do works at half strength, and their HP bar turns amber. At 25 or below it turns red, and they are too hurt to help with choices that need them. At zero they die. The captain has real health too.
- **Healing:** everyone recovers a little each leg (faster with a medic aboard). The medic's Treatment, a medkit, and MEDBAY at a station heal properly.
- **Festering wounds.** A bad wound stays open: it loses HP every leg until a medic, a medkit, the MEDBAY or rest gets the person back over the wound line, and if it is still open at the next station a wound crisis plays there. The log says each leg that it is getting worse.

- **Morale is per person.** Changes to "the crew" hit everyone, but each specialist also reacts to their own concern: the botanist to the crop's health, the engineer to hull damage (and cheers up at a finished repair), the chef to short rations, the medic to injuries and deaths, the pilot to the pace, the xenobiologist to alien finds, the diplomat to fighting, and everyone to their own wounds. Hover a morale bar to see why it moved. With only one crew member aboard, those personal concerns count half.
- **Desertion.** When someone's morale hits zero you are warned, and they will walk off at the next station unless their morale is back up to 25 by then. The morale bar and dossier say so.
- **Mutiny.** If the crew's *average* morale reaches zero, they take the ship, and the run ends.
- **The captain has no morale.** Flying solo (no crew at all) means no morale to manage and no effort tax, but also no specialists and fewer hands.
- A vacant crew slot can sometimes be filled on the way, by someone you take aboard. The ship never carries more than captain plus two crew; anyone extra rides along as a passenger. See People who come aboard.

Captains

You are the captain, and your background is what you studied before this. It stands in for a missing specialist at **half strength**: half their passive effect and half their effort contribution. If you also hire that specialist, the better of the two applies; they do not stack. Some backgrounds also bring an item or extra gold.

Background	What it gives you (as the setup card says)
Botanist	Crop water –7% · Crop +5 health at the start · tends the growbay itself
Engineer	Fabricator wear –10% · Hull starts at 105
Medic	25% of deadly outcomes become injuries · starts with a medkit
Pilot	Fuel –5% · Fuel starts at 105
Chef	Food –7% · Food starts 5 higher
Xenobiologist	Anomalies safer (stands in for a xenobiologist's choices, at half the gains) · starts with a spectrometer
Diplomat	Gold +15
Merchant	Prices –10% · Gold +30
Academic	Research: one choice previewed per station leg · starts with a gas analyzer
Veteran	An edge in combat · starts with a pressure suit

The captain also brings their own pair of hands to every job, and has real health: if the captain dies, the run is over.

The five AIs

Your AI is the voice of the whole game: the log, the bubble, the minigame briefings, the Ship's Log. Each has a personality and one passive perk.

AI	Personality	Perk
ARIA	Relentlessly positive, factually questionable. Every disaster is a growth opportunity.	Every crew member gains a little morale each leg.
MARV	Clinically depressed, 400% smarter than everyone, very tired of being right.	**Reveals hidden details**: in events, MARV flags the costliest choice with what it will cost you.
REX	Retired military. Everything is a threat, including the snack dispenser.	**Combat bonus**: the first enemy hit of every fight is stopped by point-defense.
CHIP	Corporate-issued, peppy, full of Terms of Service. Every crisis is a sponsored message.	**Prices –15%**: every market (lane traders too), station services and gold spent in station events.
AJOY	A passive-aggressive backseat driver. Technically helpful, in the most grating way possible.	**Shows what you missed**: after you choose, AJOY tells you what one of the other options would have done.

Crops and how they grow

Your crop grows from 0% to 100% through four stages: **Sprouting**, **Vegetative**, **Flowering** and **Maturing**. At 100% you can harvest it, and it starts again from a fresh planting. Its **health** is what matters for the mission: it must be above zero when you arrive.

- **Growth** is fastest when the plant gets the water it needs and someone good tends it. Being thirsty, pests, and the wrong surroundings slow it down.
- **Health** slowly recovers with good watering and falls with thirst, pests and stress. Ordinary imperfect care mostly stalls it; real neglect kills.
- **Surroundings matter.** Deep void runs cold, anomalies run hot, nebulae and the void are dim. The growbay shows these as modifiers while you are there; they pass when you move on.
- **Pests** turn up now and then once the plant is established. TREAT them in the growbay or use a crop treatment.
- A **healthy crop helps the air**: it makes some of the oxygen the crew breathes.

Crop	Character
Wheat	Reliable and predictable, with a good food yield. Recovers faster once it is thriving, but pests hit it twice as hard.
Tomatoes	The fastest food crop and a joint-best harvest, and a healthy tomato bed lifts the crew's morale a little every leg. Thirsty, hit harder by thirst, and prone to drama.
Sweet Potatoes	The hardiest: highest starting health, widest comfort zone, lowest water need of the food crops. Slow to grow and slow to bounce back from damage.
Soybeans	Pest-resistant, and recovers twice as fast for ten days after a pest outbreak is cleared. Needs steady light.
Zinnias	No food at all. Healthy zinnias lift morale every leg, a harvest gives a big morale boost instead of food, and they have the fastest cycle. Fragile, and sick zinnias make the crew miserable: hard mode.

Some crops also add a bonus to your arrival score if they arrive healthy; the report card shows it as CROP BONUS.

Items and materials

- **Items** are things: instruments (a gas analyzer, a geiger counter, a portable drill…), consumables (fuel cells, ration bricks, water and O2 canisters, medkits, hull patches, crop treatments), trade goods worth gold, curiosities, and **upgrades** (a bigger fuel tank, hull plating, a shield module…). Gear takes a cargo slot each; small consumables stack several to a slot; upgrades take no slot and work as long as they are aboard. Many items open choices in events; the hold's detail panel lists where an item is used. Some story items take no slot either: papers, tokens, blueprints, the odd keepsake.
- **Materials** are raw stock: metal, scrap, minerals, exotic matter and biocomponents. They come from mining, salvage, fights and events, and go into REPAIR and the fabricator, or get sold. Every 10 of a material takes one cargo slot.
- **The hold has limits.** Capacity depends on the trail (and one upgrade adds more). A full hold means new finds stay behind: briefings warn you before you spend the time.

Species space and stations

Every voyage crosses the space of one alien species, a different one each run, picked at launch. Your AI names it in the log as you leave Earth Orbital Station. Expect to meet that species often: its traders and pirates, its stations and its stories.

The stations change as you go. Near Earth they are human. Further out they are mixed, with people of every kind working the docks. On the Mars and Interstellar trails, some of the far stations belong to a single species, and everyone there, from the dockmaster to the vendor, is one of them. The map names such a station after its species (see The route map). On those trails the last station before your destination always belongs to the species whose space you are crossing. The Lunar trail never goes far enough for a single-species station.

Each species keeps its own customs. Getting on well with one can leave you holding a token of their regard: a writ, a chit, a receipt for something you have not bought yet. It takes no cargo slot, and while you carry it, it opens choices in that species' other encounters.

People who come aboard

Your crew is not fixed at launch. On the way you may find a survivor who needs a ride, someone at a station bar who would rather be on your ship, or a stowaway who has been hiding aboard since Earth. A stowaway only turns up once you are some way into the trip.

- If a crew place is open and nobody aboard already holds their specialty, a newcomer joins the crew in their own role, with their own name, face, skills and morale.
- Otherwise they ride as a **passenger**. Passengers hold no crew place and do no work, but they eat. The hub names them in its passenger line, and the hold's food readout counts them.
- A rescued passenger steps off at the next station where you *dock*, and may leave a thank-you behind. A station you wave off, or fly straight past, does not count: they stay aboard to the next one. A crew member who stepped down to make room for a stowaway leaves the same way.
- A stowaway you welcome who has no crew place has nowhere else to go, and rides with you to the end.
- A captain who has lost the whole crew but still has passengers aboard is not alone, and the ending says so.



A full crew and one passenger. The passenger line sits in the corner of the ship view, so the crew strip never moves.

Errands, and events that know you

Early in a voyage, someone may ask you to carry something for them: a sealed case, an overdue library book, a museum piece, and other things. If you take it on, the errand comes back over the trip a step at a time, with its last step near the end. Only one errand runs at a time. Each alien species also has an errand of its own, which is only offered on a voyage through its space: the species you cross decides which ones can come your way.

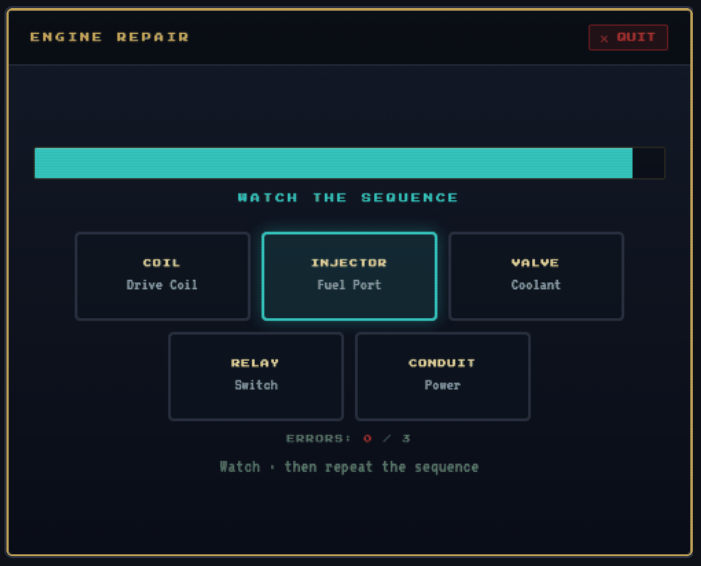
Some events also know who you are: what your captain studied, the crop you carry, the AI you picked, how far along the route you are, and even a ship you lost on an earlier voyage. Each trail has scenes of its own, and the Lunar and Mars trails have landmarks along the way.

The minigames

Hands-on jobs open as minigames. Each starts on a briefing with your AI's take and the rules, then **BEGIN**. The result is graded **EXCELLENT**, **ADEQUATE** or **INSUFFICIENT**, and a better grade means a better reward. When an event launches a minigame, the briefing lists exactly what each grade earns.



Scan. Keep the cursor on the wandering blip until the lock bar fills.



Engine restart. Watch the sequence, then click the parts in the same order. Too many wrong clicks and the drive limps.



Solar brace. Hold to brace, release to vent heat; let go when the screen flashes orange before a surge.



Medical triage. Scan, reveal each wound, then match the treatment to its color, in the bracketed order.



Forage. Click creatures fast, pick plants at their gold glow, and never click a decoy.



Hull breach. Drag a patch onto a hole, then click it to weld. Against a real clock.



Salvage. Hold on a drifting item to tractor it and drag it into the bay. Long drags burn energy.



Engine-room repair (the REPAIR job). Find and fix four faults, then kill the lights.



Mining. Click rocks to crack them. Every click costs beam energy.



Combat. Turn-based: pick an action, then they shoot back.

Minigame	Where it comes from	How to play
Scan	The SCAN job; some events	Move the cursor onto the blinking blip on the radar and hold it there until the lock bar fills. The blip wanders: stay with it. Harder on longer trails.
Engine repair	The REPAIR job (the engine room); engine-fault events (the restart sequence)	*Engine room:* click each of the four highlighted faults and fix it: reset the four tripped breakers; flip the ignition switches to match the laminated reference card; hold the bleed valve until the gauge needle sits in the green (overshoot, and you must release and try again); reconnect the three fuel lines. Then turn off the lights to unlock the bridge. RETURN TO BRIDGE before the end abandons the repair. *Restart sequence:* the components light up in order; click them back in the same order from memory. A few wrong clicks and the drive goes to limp mode.
Medical	Injury events	Turn on the scanner, then click each unknown wound to see what it needs. Apply the treatment that matches its color, in the order the brackets mark. The wrong treatment hurts the patient, and the patient will not wait forever.
Forage	The FORAGE / GATHER / HUNT / SCAVENGE job	Targets pop out of cover at fixed spots. Animals vanish in about a second: click fast. Plants ripen: wait for the gold glow for the best yield, but past it they spoil (poor food and a morale hit). Decoys are eye-marked or oddly tinted, and clicking one poisons someone aboard. DEPART keeps your haul early; decoys get more common as the timer runs down.
Brace	Solar-storm events	Hold the mouse button to brace: the shield holds, but heat builds. Release to vent. If heat hits 100% the brace fries. The screen flares orange about a second before a surge: release during the strike or take the hit. Shield remaining at the end is your score.
Breach	Hull-breach events	Drag a patch from the tray onto a hole until it snaps in, then click it to weld (it glows orange when it holds). Drop a patch outside a hole and it falls; pick it up off the floor. One patch at a time. The countdown runs on a real clock; an engineer buys you a little more time per breach.
Salvage	The SALVAGE job; some events	Click and hold on a drifting item to grab it with the tractor beam, drag it into the glowing cargo bay (or the hatch, inside a derelict) and release. The beam drains energy while on, so short

		drags are cheaper. Items that drift away are gone. Crates open on the results screen; sparking junk is worthless.
Mining	The MINE job; some events	With crew aboard, first pick one to assist (each tilts the haul toward a different material), then BEGIN MINING. Click asteroids to crack them for materials; every click, hit or miss, drains the beam's energy. Big and rich rocks give more, hollow ones give nothing, and hits in a row build a chain. The run ends when energy or time runs out, or when you EXIT MINING. Then MINE AGAIN (another day) or RETURN TO SHIP.
Combat	Hostile encounters, when a choice starts a fight	On your turn: **FIRE LASER** and click their ship to aim (or FIRE AT CENTER); **EVASIVE MANEUVERS** to halve the next hit; **HULL PATCH** once, with an engineer, when the hull is below half; **BOOST SHIELDS** once, with a Shield Module, to block the next hit; **ATTEMPT ESCAPE** to charge the FTL drive for a few turns while they keep shooting, then jump away for some fuel. A botanist or chef can offer a tribute on the first turn. If they surrender you can accept (less loot, better for morale) or finish them (full loot, and it is remembered). Win and you collect loot from their wreck.

> **Quitting a minigame** once it has begun keeps nothing and still spends the time. In an emergency, quitting means the worst result lands. Closing on the briefing, before BEGIN, is always free.

Mishaps, a cracked hull, and running dry

Mishaps. Not everything is an event you choose. On a travel leg something can simply happen: a pinhole breach someone plugs with chewing gum, fuel spilled in a corridor, a clogged scrubber, a grease fire in the galley, the captain slipping on a wet deck. The cruise stops on a popup you have to acknowledge (it has no ✕, so you can't miss it), and the Ship's Log records it in red. On Lunar most are small. Elsewhere they can cost real hull, oxygen, fuel, supplies or health; they come more often at faster paces, and they hit harder on Mars and hardest on Interstellar. The right person aboard turns a bad one into a small one: the engineer for the hull, air and tanks, the pilot for the fuel, the chef for the galley and the stores, the botanist for the growbay. A crew specialist always comes through; a captain of that background, or a wounded specialist, half the time. Not every run has a mishap, a run can have several, but never on two legs in a row, and never on the leg into the destination.

A cracked hull cracks further. Below 50 hull, mishaps come more often and lean towards the hull, and a breach costs air as well as hull. Below 25 they are likely, and the severe ones much likelier, so a neglected hull can spiral into suffocation or a breach. REPAIR, the DRYDOCK and hull plating are how a careful captain stays out of that spiral.



A mishap. It has to be acknowledged; the Ship's Log keeps it in red.

Adrift. Running out of fuel no longer ends the run on the spot. The leg that empties the tank still arrives, so a station or a trader there can still save you. Try to fly on with nothing in the tank and the **Adrift** scene opens instead:

- **Burn the hull plating**: thrust from your own skin, at a real cost in hull.
- **Crack the water**: four days at the electrolysis rig turns a third of the tanks into fuel.
- **Trigger the emergency beacon**, if you carry one: someone always answers. A friendly hauler, a recovery tug with an invoice, or people who refuel you at gunpoint and take half your gold.
- **Call for help and wait**: days pass, and every day costs food, water and air. Someone may come (and want paying), the wrong someone may come, or nobody does, and you are back where you started, ten days poorer.
- **Shut it all down**: the end of the voyage.

The tooltips show each option's days, costs and odds before you pick, and warn in red when your food, water or air won't last the wait. Shutting it all down is the Adrift ending. Run out of food, water or air while you wait, and that is the Adrift ending too.



Adrift: out of fuel short of the destination, with a few hard ways on.

The captain in harm's way. Some hard scenes put the captain personally on the line: an EVA, a boarding, volunteering for the dangerous part. And a **festering wound** (the kind that stays open) gets worse every leg until it is treated, the captain's included. Losing the captain is a real way to lose.

Deadlines and arriving late

Each trail has a deadline: the day the colony expects you. The day counter shows the days left, and ORDERS projects your arrival day at the pace you have set. If that projection drifts past the deadline, you are behind schedule, and the trail may let you know.

Being late never ends a run. When the deadline passes, your AI says so once and the ship flies on. You still arrive and still get your ending; the day counter switches to counting the days late, the report says how many, and every late day comes off your score. Arriving early adds to it.

How a run can end

Arrival. Reach the destination and the run is scored. The score weighs how much of your crew made it, how well-stocked the ship is (fuel, food, water, hull and morale), and how early or late you are, plus any extras such as soil surveys and a crop bonus. The score sorts the arrival into one of four tiers, from a triumphant **MADE IT** down to an arrival nobody is quite sure about, each with its own endings to discover (twelve arrival endings in all, counting the two for a dead crop). Each trail has its own scale, so a good Interstellar arrival is judged against the Interstellar trail. And if the crop is dead when you arrive, none of that matters: the crew made it, the mission did not, and the game will let you sit with that.

Ending early. A run can also end on the way:

- the ship is **adrift** with no fuel and you shut it all down, or the supplies run out while you wait (see Adrift);
- the **hull** fails;
- the **captain** dies (whether or not anyone else is left aboard);
- the crew's average **morale** hits zero and they mutiny;
- too many days with no **food**, no **water** or no **oxygen**;
- the **crop** dies.

Each has its own screen: nine in all, since food, water and air each get one, and a lost captain reads differently with or without anyone left aboard.

Some of these leave a door open. When the crop dies you can choose to keep flying, though your arrival can then only be the hollow kind. An empty tank is not the end by itself: the Adrift scene gives you ways on, and an emergency beacon makes sure someone answers. If the whole crew is lost but the captain lives, the captain flies on alone (or with whatever passengers are still aboard).

Tips for surviving

Every trail

- **The crop comes first.** Check the crop card every stop. Keep its water share at or just above what it needs, TREAT pests promptly, and harvest when it is ready: the food (or the morale) is yours, and the plant starts again.
- **Read the tooltips.** Every cost the game knows about is on a tooltip before you commit: days, supplies, morale, odds, and what happens if you fall short.
- **Mind the pips.** Heavy jobs tax every crew member's morale. A few big jobs in a row can leave the crew miserable even when everything went well.
- **Dock at stations.** Docking recharges active skills, stations have the biggest market, and only stations will heal your crew or patch your hull for gold. Waving off is free in time but expensive in options.
- **Salvage pays.** Derelicts and wreckage are where steady gold and materials come from, and gold buys medbays, drydocks and upgrades.
- **Use your specialists' active skills.** They recharge at the next station, so an unused one is wasted.
- **Watch the ORDERS projection.** It tells you the day you will arrive and the day each supply runs out at your current settings.
- **SCAN before a fork** when you can spare the day.
- **Keep the hull above 50.** Below that, mishaps come more often and hit the hull harder; below 25 they snowball.
- **Treat a festering wound before you fly on.** It loses HP every leg, the captain's too.
- **Do not burn Plaid forever.** Fast is tempting; the engines, the fuel and the crew all pay for it.

Lunar: the gentle trail

- Lunar forgives a lot. Use it to learn the screens, try the minigames and get to know a crew.
- Stations are plentiful, so spend gold freely on repairs and healing.
- With a short trail, every lost day is a bigger share of the deadline: do not rest more than you need to.

Mars: the thirsty trail

- You leave short on water and food, and events here ask much more of both. Water is the supply most likely to bite.
- Do not over-water the crop "to be safe": give it what it needs and keep the rest for the crew. Ice from mining asteroids and water canisters are worth the time.
- A chef, sensible rations and regular harvests keep the larder going; a food crop pays for itself here.
- Trouble gets worse as the trail goes on. Reach the back half of the trip with supplies in hand, not scraping by.

Interstellar: the long trail

- It is a long way through stranger places: voids, anomalies, long stretches between stations. Plan for long gaps between resupplies.
- The crew starts less cheerful. Keep morale healthy early; it is much harder to rebuild later.
- The hardest events wait for the final stretch. Bank fuel, hull and spare consumables for the end.
- The fabricator earns its keep here: keep some materials and service it at stations.
- Fuel matters over this distance. A pilot, sensible engine settings and a stock of fuel cells keep you from drifting, and an emergency beacon makes drifting survivable.

- Mishaps come about as often as on Mars but hit much harder: about a third of them are severe. An engineer aboard pays for itself.

PART 2 · BEHIND THE CURTAIN

How the game works underneath: the moving parts, how they talk to each other, and why the ship does what it does.

Spoilers past this point. Part 2 opens the hood: how events are picked, how the score is worked out, what the crew is quietly keeping track of. If you'd rather discover all that the hard way (the traditional way, in space), close this part now and come back after a few voyages. Nothing here is required to play. It is, however, extremely satisfying to read after a run went sideways.

This part describes concepts, not code. It explains each system in plain words and draws the connections between them. Where an exact number matters to understanding something, it gets a mention. Otherwise you'll read "a little", "a lot" and "much more likely", which is how the ship's crew would describe it too.

The big picture

Trans-plant is one long-running story about one ship. Everything the game knows about that story lives in a single place, **the run's state**. Every other part either reads it, changes it, or both.



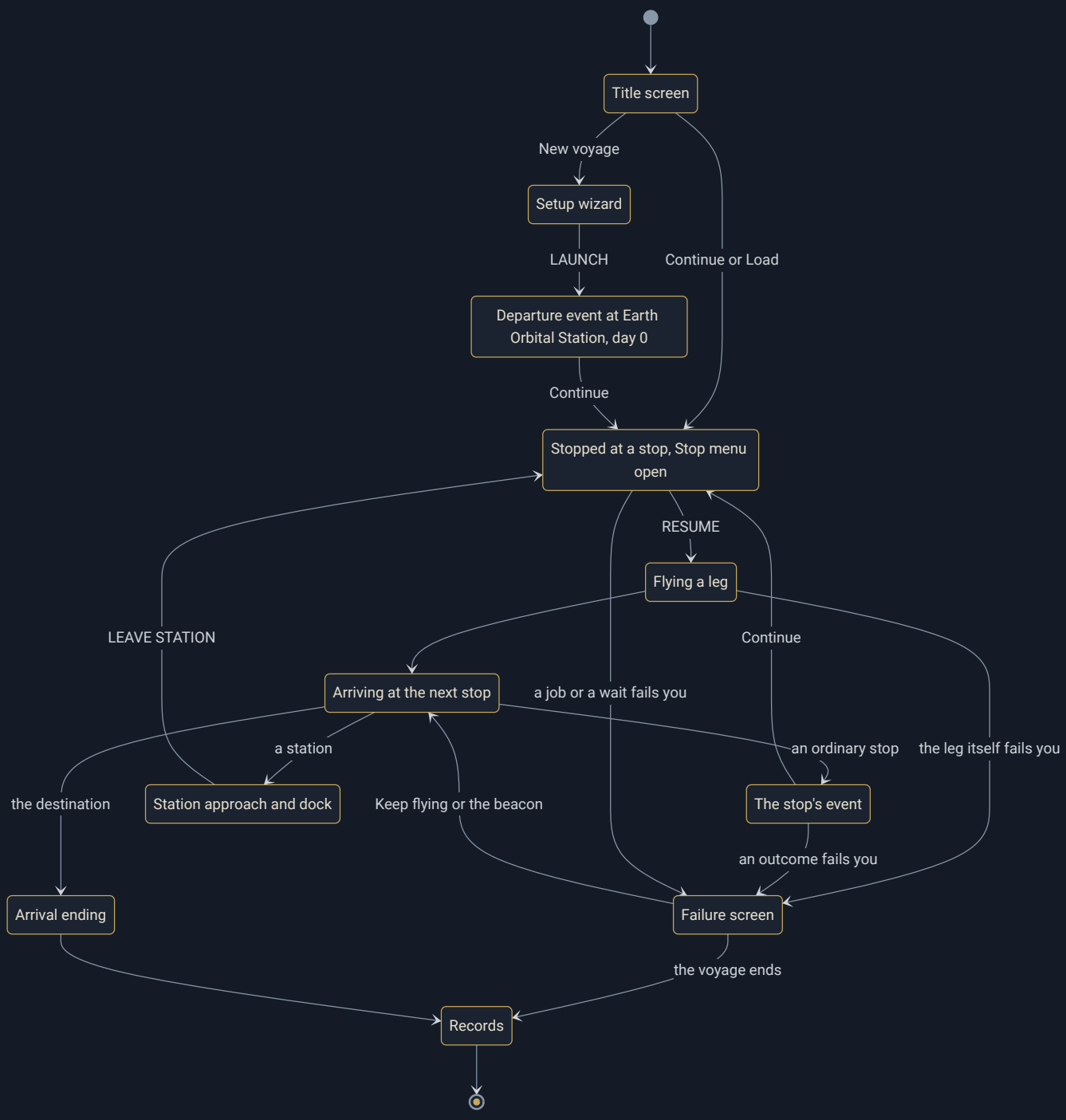
The game's moving parts. The run's state sits in the middle; almost every arrow goes into it or comes out of it.

- **The content library** is the game's data: trails, events, crew, captains, AI companions, crops, items, materials, recipes, endings, failures and a great many flavor lines. It is loaded once when the game starts and never changes during a run.
- **The map generator** builds a fresh route for the chosen trail at launch, so no two voyages share a map. It also decides who lives at each station and which stations keep a back room.
- **The run's state** is the voyage itself: supplies, crew, crop, cargo, the map and where you are on it, the day, earlier choices, and the log.
- **Screens** show the state and turn your clicks into changes: the cruise hub and Stop menu, encounters, Orders, the growbay, the crew dossier, the cargo hold, the market, the fabricator, the map.

- **The event system** decides what happens at each stop, runs the encounter you play through, and applies its outcome.
- **The ship's daily clock** charges each day that passes: food, water and air drawn, morale drifting, the crop growing, the deadline getting closer.
- **Minigames and jobs** are the hands-on work: scanning, mining, salvaging, foraging, repairs, medical emergencies and so on. Their result decides what lands.
- **Combat** is the ship-to-ship fight some events can start. It keeps its own little scoreboard during the fight, then hands the net result back.
- **The failure check** runs after anything changes, and ends the voyage the moment one of the ways to lose comes true.
- **Endings** write up the voyage: an arrival tier and score, or a failure card.
- **Records** remember finished voyages and each destination's best, separately from any save.
- **Saving** snapshots the state at safe moments, so Continue and Load can bring the voyage back.
- **The AI companion** you picked gives the whole thing a voice: event lines, the log, tips, warnings and a few real perks.

A run from start to finish

A voyage moves through a small number of states. Most of the time you are either *stopped at a stop* (the Stop menu open) or *in an encounter*. Flying a leg is instant: pressing RESUME pays the leg's cost and lands you at the next stop.



The lifecycle of a voyage. Accepting a failure that offers a way on carries on from where it struck: a stop it interrupted on the way in still plays; otherwise you are back at the hub.

Setup and the departure

The setup wizard collects the trail, the AI, the captain's background, up to two crew, the crop and the ship's name (left blank, the ship draws one). LAUNCH turns those choices into a fresh state: the trail's starting supplies, everyone at full health, each crew member at the trail's starting morale, the captain's starting bonus, a small default loadout, and a freshly generated map. Just before the map is built, LAUNCH rolls the **voyage species**: one of the six alien species, whose space this run crosses (see Species space). About a third of voyages also quietly load a **stowaway**, who stays hidden for now.

Then the departure plays: the AI names the voyage species in the log, and one launch event plays at Earth Orbital Station on day 0. Its Continue opens the hub, stopped at the station, with the Stop menu up, exactly like every later stop.

The loop: stop, resume, arrive

After every event the ship is *stopped*, and the Stop menu is already open. RESUME is what flies. Before it travels, it checks a few things:



What the RESUME button actually does. At a fork, the fork's own event decides the branch; after that, RESUME has exactly one place to go.

A **fork** is a stop of its own. Arriving there plays a route-decision event, and the choice you make in it sets which branch you're on. From then on the map shows one way forward and RESUME takes it.

The end

Reaching the destination ends the voyage at once with an arrival ending. Anything that empties the ship of a way to keep going ends it with a failure. Two failures offer a way on: a **dead crop** (Keep flying) and an **empty fuel tank** when you carry an emergency beacon. Choosing either un-ends the voyage and plays the stop that the failure interrupted on the way in, so it can't be skipped for free.

Saving, Continue and Load

The game keeps one **autosave** (what Continue resumes) plus three manual slots in the pause menu (SAVE and LOAD). The autosave is written:

- every time the hub is redrawn, which is every safe moment between actions;
- when you enter an event, and at every step deeper into its layers;
- the moment an outcome lands (so a reload can't offer the same choice again);
- with a special marker the moment a job begins, a fight changes, or a failure that offers a way on appears.

A manual SAVE is refused in the middle of an encounter or a job; the pause menu says why. A voyage that ended for good clears the autosave. A finished voyage can't be resurrected.



What a reload brings back. The rule behind all of it: a reload is never an undo button.

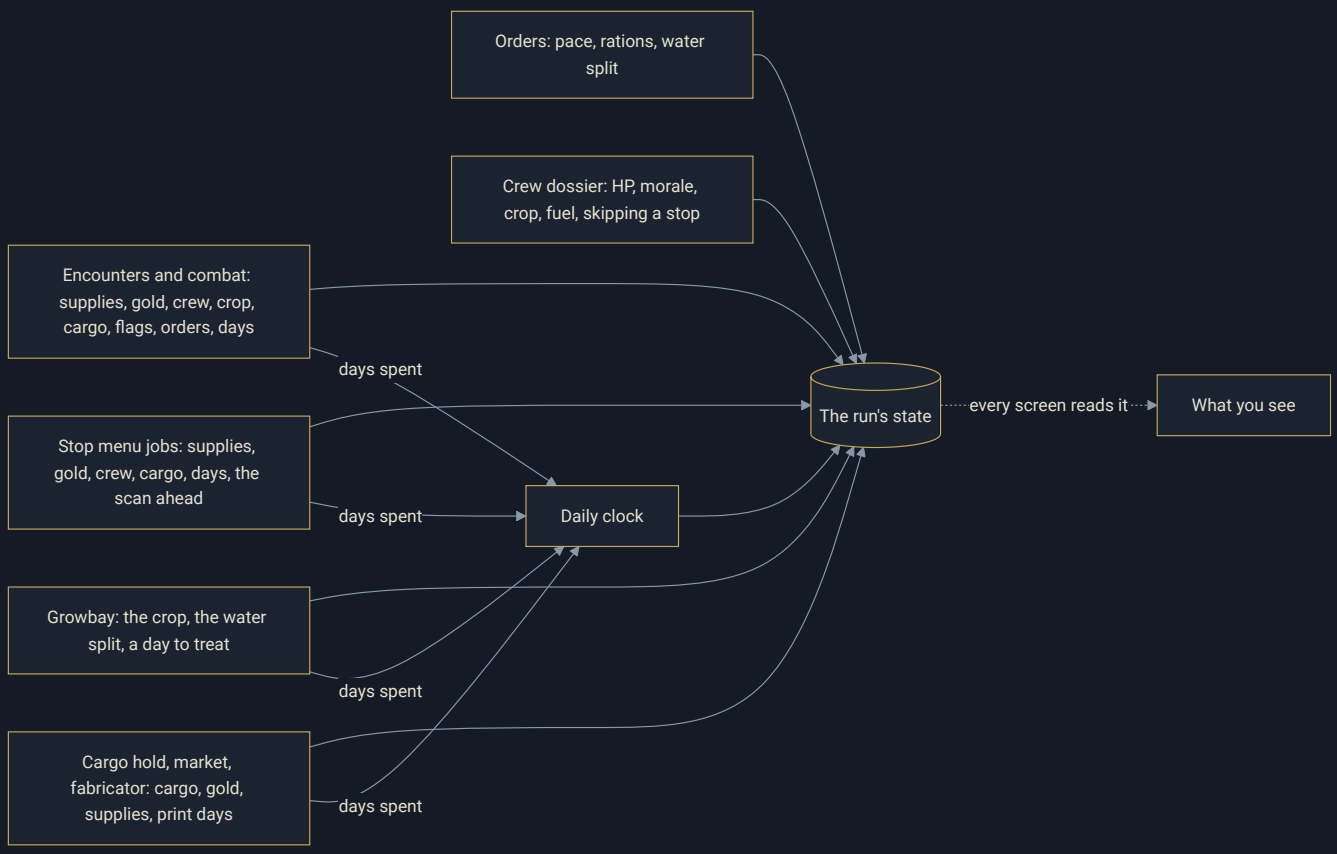
A reload mid-fight is generous only if the fight was already decided. If you had already won (a kill, a surrender, their retreat, your jump), that result lands. If you were in the defeat beat, the defeat lands. Otherwise the game plays out ATTEMPT ESCAPE for you, with the enemy firing at the drive the whole way, which is never better than playing on.

What the game remembers

Here is the run's state in plain terms. Everything below is saved with the voyage.

PART	WHAT IT HOLDS
Supplies	FUEL FOOD H2O O2 HULL, each 0 to its ceiling (some upgrades and captain bonuses raise a ceiling), plus GOLD.
Crew	Who is aboard and in which role, each person's HP (the captain's too), each crew member's own morale and the reason it last moved, who has been warned at zero morale, open wounds waiting for treatment, active-skill charges, newcomers' identities (a joined stowaway or rescue keeps their own name and face), passengers.
Crop	Which crop, its growth and health, pests, who tends it, days since planting, whether the crew talks to it.
Cargo	Items (each trade good with its own value), materials, which are new, and the hold's capacity.
Flags	Facts about choices you made and things that happened: "chose the beta branch", "executed a surrendered pirate", "met the anomaly", and well over a hundred more.
Map and position	Every stop and its connections, who lives at each station and which keep a back room, the voyage species, where you are, where you've been, and any events a SCAN locked in ahead.
Day and deadline	Days elapsed, the trail's deadline, and the fraction of a day carried from leg to leg.
Standing orders	Pace, rations and the water split between crop and crew.
Ship systems	Fabricator wear (and a print in progress), drive heat from PLAID burns.
Event memory	Which one-time events have happened, when each repeatable event last fired, the hard-event streak, the post-combat window, which NPC each event rolled.
The log	The Ship's Log journal, plus what the AI has already said (tips, callbacks, warnings) so it doesn't repeat itself.

Every screen reads the state. The diagram shows which screens *change* which parts.



Who writes what. Encounters can touch almost anything; Orders only ever changes the orders. Days never move on their own: every screen that spends time hands it to the daily clock.

A few notes on the arrows. The Stop menu changes *position* through SCAN, which locks in what waits ahead. The crew dossier changes it through the pilot's Push Engines, which flies past the next stop. The growbay changes the orders because its water slider and the one in Orders are the same setting.

What isn't saved

Only the moment-to-moment bits are left out: text halfway through typing, the AI's speech bubble, a minigame or fight in progress (replaced by the markers described above), a wait in progress (a reload calls it off), and settings such as text speed and tips, which belong to you rather than to the voyage. The art seed *is* saved, so a reload shows the same backdrops and the same faces.

How an event happens

Every stop gets at most one event of its own. What it is depends on where you are, how the voyage is going, and what you did earlier.

What plays when you arrive



The arrival sequence. A wound crisis even beats Push Engines: you can't outrun an infection.

Stations

A station stop is a little visit rather than a single scene. On arrival, anyone who was warned at zero morale and hasn't recovered walks off, before anything else happens. Passengers wait for a **berth**: they step off when the ship docks (and a rescued rider's family pays up then), or on arrival at a station stop that skips the approach (a wound crisis, or a quiet stop with nothing drawn). A station you wave off, or one Push Engines flies past, is no berth, so the riders stay aboard to the next one. Then:

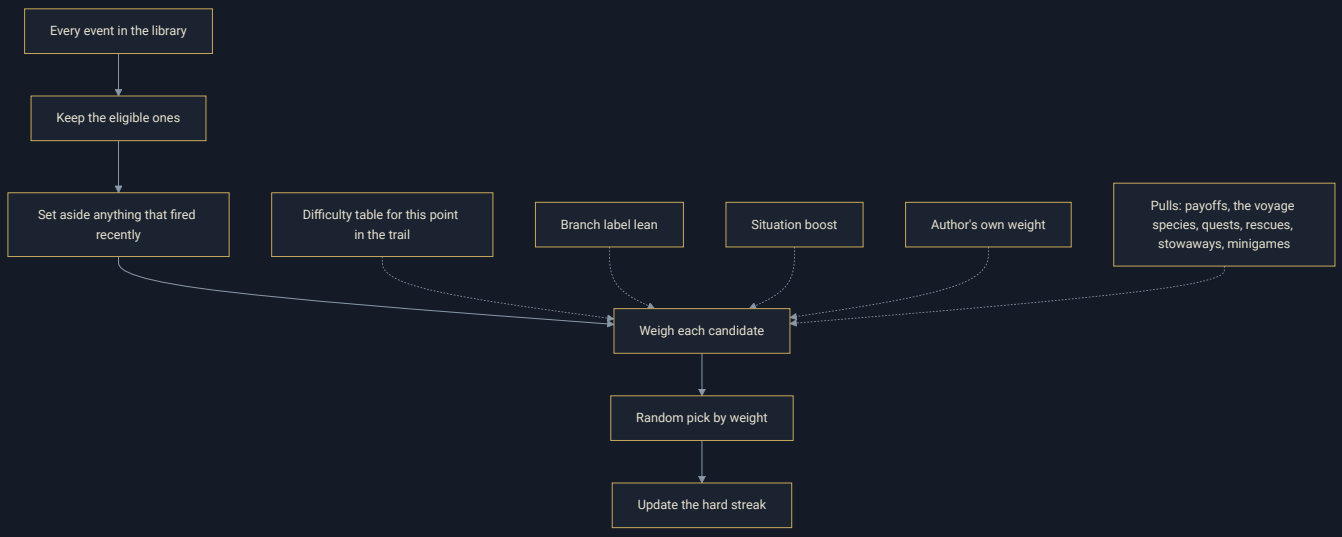

Arrive at a station
Is the drawn event a station scene?
yes
no, a shipboard or open-space one
It plays first
The approach: HAIL, DOCK, or wave off
DOCK, maybe a parking fee
wave off
Crew skills recharge, passengers step off
a docking scene
an in-person scene, or none
It plays as you dock
The dock: a room for the scene, and Market Row
Services closed for this stop, no time spent
LEAVE STATION
The hub: station services, and any back room, open if you docked

A station visit. Sometimes the dockmaster hails you before you can, and sometimes docking costs a small fee. A room you've finished stays finished for the visit.

The pool: which events can appear

An event is *eligible* at a stop only if every one of its conditions holds right now:

- **Stop type.** Events are written for planets, stations, asteroid fields, derelicts, nebulae, the void, anomalies, or anywhere. Forks are special: only route-decision events can appear there, because the fork's event has to pick a branch.
- **Trail.** Some events belong to one or two trails only, which gives each trail its own texture. The Lunar and Mars trails have landmarks of their own; the Mars landings wait until most of the route is flown.
- **Who this run is.** An event can be written for one captain background, one crop or one AI, and only enters the pool on a run that has it.
- **Route progress.** An event can wait for a share of the route to be flown, so it lands mid-run or near arrival.
- **Your own history.** A few events need a lost ship among your past voyages (a run in the records that ended in failure), and name its ship, captain, crop, fate and days.
- **Where the species are.** An alien species' events never fire in the human-only stretch of the route, and at a station only if that species lives there or the station is mixed. A few are written for a station wholly that species' own, and fire only there (see Species space).
- **Planet biome.** A scene written for one kind of world only fires on a planet of that kind (the same biome its art shows).
- **Situations.** Many events need a situation to be true: low on a supply, low crew morale, the start, middle or end of the journey, running behind schedule, a strained drive, a sick or pest-ridden crop, a dead crop you kept flying with, someone injured, a death earlier in the voyage, flying solo, the few stops after a fight, a worn or broken fabricator, a full or empty hold, a hidden stowaway past the first fifth of the trip, and two good-fortune ones: a thriving crop (in fine health, well grown and free of pests) and a purse worth somebody's trouble.
- **Story flags.** A follow-up only joins the pool once the earlier choice that sets it up has happened. Some events also stand aside while a flag is set, which is how a quest keeps others out of its way (see Quests). A follow-up about a welcomed stowaway also needs them still aboard.
- **What you carry.** A delivery or payoff event only fires while the story object is still in your hold. Jettison it and the payoff never comes. The one exception runs the other way: the Borrowed Charm can't be jettisoned at all, so its debt always comes due.
- **Who is aboard.** A scene where a crew member speaks needs someone who can say it; a scene about a particular role needs that role aboard as crew; a crop story needs a living crop.
- **Once only.** Some events happen once per voyage. The departure and the halfway moment are one-of-a-kind too.



Drawing from the pool. Everything eligible stays possible; the weights only make some far more likely than others.

Difficulty shifts later in a trail

Every event is tagged easy, medium, hard or ultra. Each trail has a difficulty table for its early stops. Past a point in the journey, the table slides steadily toward a harsher one by the time you reach the destination. The shift only changes the odds; it never adds or removes events. On the Interstellar trail, the ultra tier shows up only on the late legs.

Repeat cooldown and families

Most events happen once a run: every event that tells one particular story (a named ship, a running joke, a discovery, a set-piece in a species' space). Routine situations can repeat: low supplies, pests, wounds, breakdowns, station services, the everyday trader and pirate. An event that can repeat sits out for the next four stops after it fires. Some events belong to a **family**, such as the traders or the pirates. A family cools down as one, so meeting one pirate crew keeps all the pirates away for a while. A family's named ships (the Museum, the closing-down sale, the learner pirate) are met once a run; the rest of the family takes over their odds, so traders and pirates turn up as often as ever. Within a family, which face you meet at a stop is fixed for that stop, so TRADE and a WAIT at the same stop bring the same trader. If every eligible event is cooling down, the stop still gets one, the one drawn longest ago, rather than going empty. Events locked in by a SCAN count as drawn for this purpose, so a scan can't line up the same scene twice in a row.

The hard-streak relief

The game counts hard (and ultra) events in a row. After a run of several, it eases off: the next couple of draws lean heavily toward easy and medium, and the late-trail shift is suspended for them. It is a pity timer for death spirals. A SCAN's look ahead doesn't count toward the streak.

Situation events

When a situation is active (low water, a sick crop, an injury), the events written for it get a large boost. They become likely, not guaranteed, so ordinary content still turns up during a crisis. A situation's events share one boost between them, so a well-stocked situation gives you *variety*, not more crisis scenes. The positional situations (start and end of

the journey) only get a mild bump. Branch labels add their own lean on top: a hostile branch favors fights, a scientific one favors science, a safe one favors easy and peaceful scenes. More on labels in The map.

Pulls

Five hundred events is a big pool, and an earned consequence must not drown in it. So, on top of the difficulty table, the branch lean and the situation boost, some kinds of event get a **pull**, a fixed extra weight:

- **Payoffs.** An event gated on a flag or an item the run holds draws four times as often.
- **The voyage species.** Its events draw three times as often, and an event written for a station wholly that species' own draws six times as often there, so it is usually what plays when you reach one.
- **Quests.** A quest's next step, once it is due, is all but certain to be the next thing that happens.
- **Rescues.** An event that can bring someone aboard draws twice as often, and four times once the crew is short-handed (smaller than at launch, or someone has died and a crew place is still open). A captain who set out alone isn't short of anyone. A rescue that has already happened stops pulling.
- **Stowaways.** Once a hidden stowaway's reveal window is open, the events that find them draw three times as often.
- **Minigames.** An event with a choice that launches a minigame draws half again as often, so the hands-on moments don't thin out as the pool grows.

The two good-fortune situations get only the mild positional bump, never the crisis boost: a run that is going well shouldn't spend every stop being congratulated.

Quests

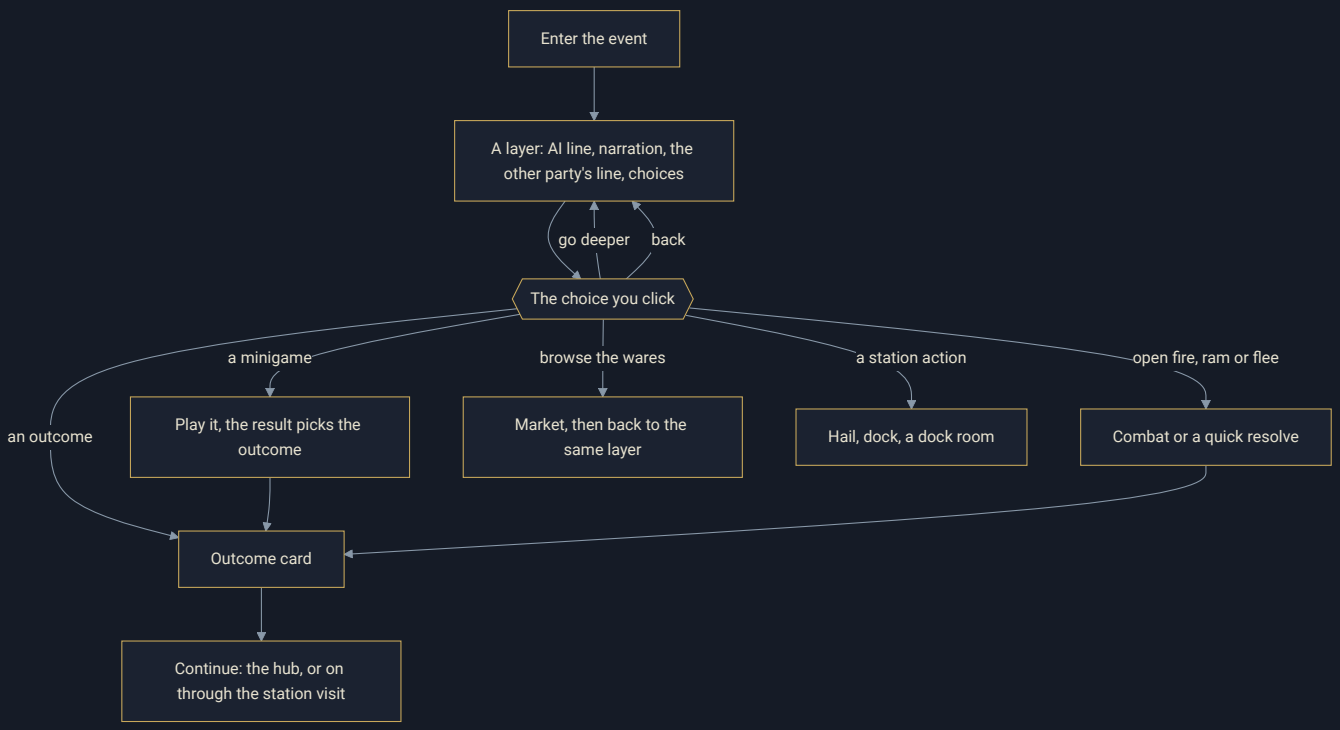
Eleven quest chains run across a whole voyage, five steps each. Each starts with an opener near the start of the trip, which hands you something to carry. Its later steps each wait for the step before and for a share of the route to be flown, and the last plays near the end. One quest runs at a time: while one is under way, every other opener stands aside.

- **Five can turn up on any run:** a landing deed, a locked case, a packet of heirloom seeds, an overdue library book, a museum's bone. Their steps wait for roughly a quarter, then nearly half, two thirds and four fifths of the route.
- **Six belong to one alien species each** and are offered only when the run crosses that species' space: Earth's Place (insect), the Witness Zircon (rock), the Census Return (water), the Petition of Majority (reptile), the Next Model (robot) and the Settled Claim (unknown). Their openers fire in the human stretch near Earth, so a human hands the job over; the second step still happens among humans, and the last three in the species' space (past half the route, then two thirds and four fifths). On Mars and Interstellar the last station before arrival always belongs to that species. The species' standing token opens the best way through.
- **Odds.** A species quest's opener draws at more than twice an older opener's weight, so about a third of quest offers are the species quest, and about half of all runs are offered a quest of some kind, as before.

Other events can wait on a quest too; The Collector, for one, stays away while the bone quest is under way, so a salvaged bone can still bring him later.

Inside an event

Simple events are one screen: a body of text and a few choices, each leading to an outcome. Bigger ones are built from **layers**, like rooms you walk between. Each layer has its own narration, a line from your AI, maybe a line from the person you're talking to, and its own choices. Choices can lead deeper, go back, or end the event with an outcome.



The shape of an event. Revisiting a layer can show different text; an action used up at a stop stays used up there.

The person across from you is rolled when the event starts (species, quirks, a name) and remembered, so revisits and reloads meet the same face. Many events also have alternate openings, rotated through the voyage, so a repeat reads differently.

Gates and locks

A choice can need a crew role, an item, gold, an earlier story flag, a living (or ailing) crop, a working fabricator, or room in the hold for what it grants. A locked choice stays visible and lists everything you're missing ("Need: Diplomat + 20g"). Story-gated choices just say so. A role you have can still be unable to help: someone hurt too badly can't answer a crew-gated choice, and the patient in a wound crisis can't treat themselves.

A choice can also show only while an item is *gone*, so the scene notices what you threw away. A **blueprint** works like an upgrade for the fabricator: carrying it unlocks a recipe.

The one stand-in: a captain trained as a xenobiologist can take a xenobiologist's choice when there isn't one aboard, at half the gains.

Previews

Every choice's tooltip tells everyone how long it takes and what it costs. Beyond that, some choices show a one-line preview of what they'll do:

- **Posted terms.** Some choices simply say what they do, such as a station botanist's advice.
- **MARV's read.** With MARV aboard, the costliest choice on screen shows its costs. ("The cost of accuracy is knowing how bad it is.")
- **A perfect SCAN.** At a stop you scanned perfectly, every choice shows its outcome.

- **Instruments.** A medical scanner shows every choice's outcome in a wound crisis; a Geiger counter flags every choice that can hurt the crew.
- **The Academic captain's research.** One choice's full outcome, once per station leg.

Previews are worked out by the same rules the outcome uses, so a preview never promises a number the ship would change (an engineer's better repairs, a recycler's saved air, the trail's supply bite).

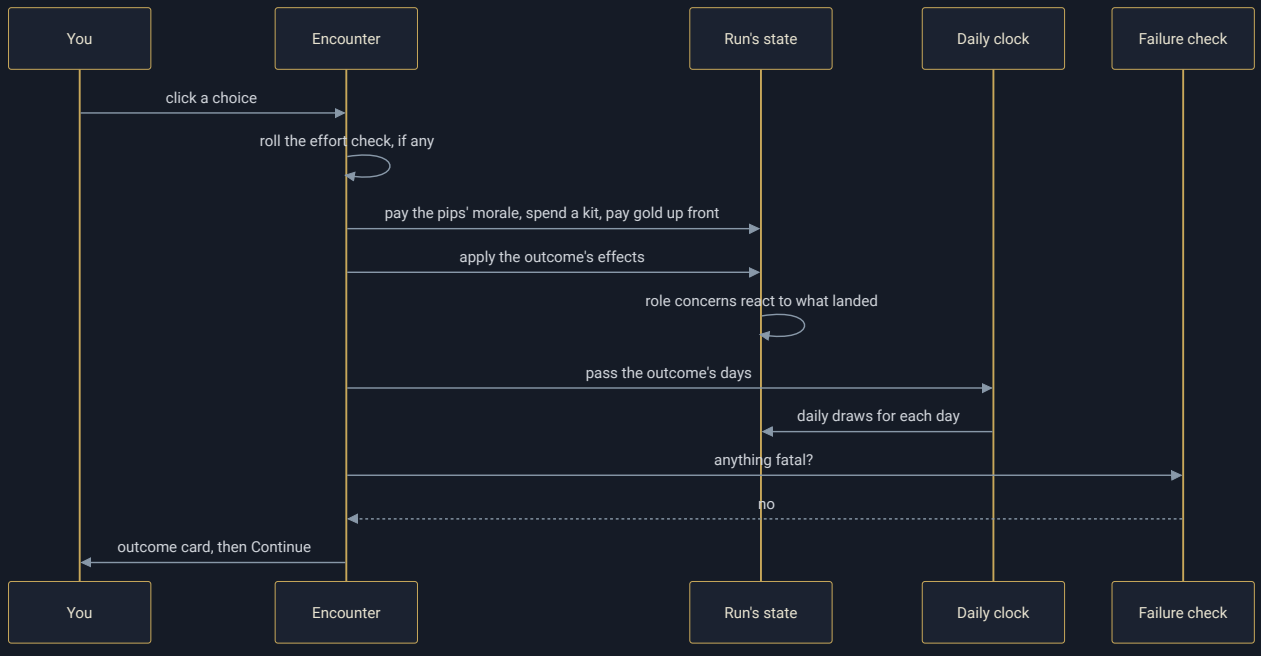
Effort checks and pips

Many choices carry **effort pips** (●●●○○). Each pip level costs every crew member some morale the moment you commit, and the cost climbs steeply as the pips go up. With no crew aboard, there is no morale to tax.

Hands-on jobs (a wrench job, a calibration, a negotiation that might wobble) also have a **hands check** (the choice shows `2 days · 90% clean`, and its tooltip names who's helping and what missing help would change; the dots are the separate effort cost). The hands you bring are the captain's own hands, plus each listed specialist at their full value (half if wounded, half for a captain of that background), plus general help from everyone else, plus the right tool from the hold. The button shows the odds. Short-handed, you get a costlier version: thinner gains, extra days, and the job's own setback (or, for some jobs, a different outcome written for the botched attempt). Even a perfect crew keeps a little risk. Some repair jobs also scale their *time* by effort: a crew that can't contribute much takes up to twice as long, and a strong one finishes faster. A one-use kit is spent only when it actually changes the result.

What an outcome can change

An outcome can move any supply or gold (as a fixed amount or a share of what you have), morale (for the whole crew or one person), crop health, items and materials (as far as the hold has room), fabricator wear or repair, pests, standing orders, flags, and people: an injury, a bad wound, a death risk, healing, someone joining, a stowaway's fate. It also takes time, from no time at all up to a couple of weeks, and that time passes through the daily clock like any other day.



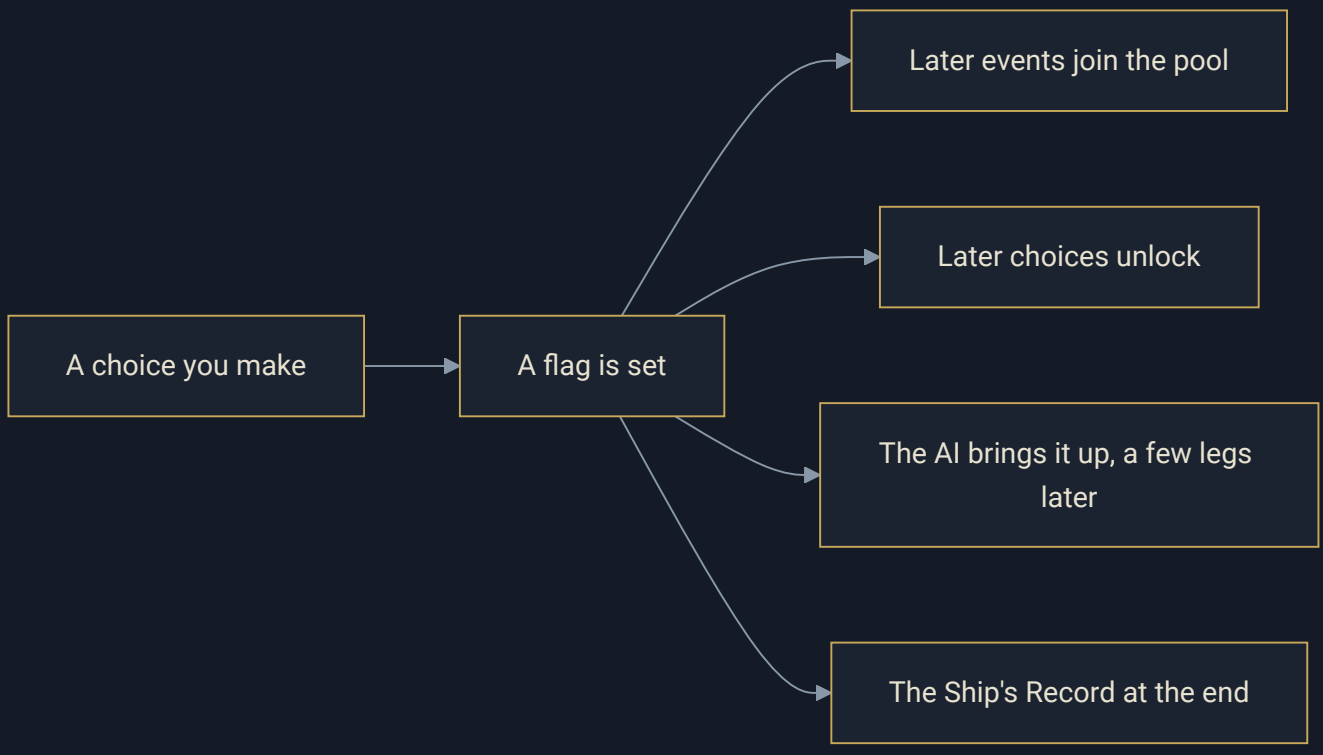
Resolving a choice, in order. If the failure check says yes, the end screen replaces the outcome card.

Severity: injury, bad wound, death risk

Harm to people comes in three grades. An **injury** drops someone below the wound line. A **bad wound** does the same and stays open: it loses HP every leg, and unless it's treated before the next station, a wound crisis fires there. A **death risk** kills, unless a medic's training turns it into a bad injury (a crew medic often, a medic captain less often, neither can save themselves), or, failing that, a carried medkit sometimes does, and is used up.

Flags that come back

Choices leave flags behind, and flags echo. The diagram shows the four ways one decision can return.



One choice, four echoes. Callbacks wait at least two legs, and come at most every other stop.

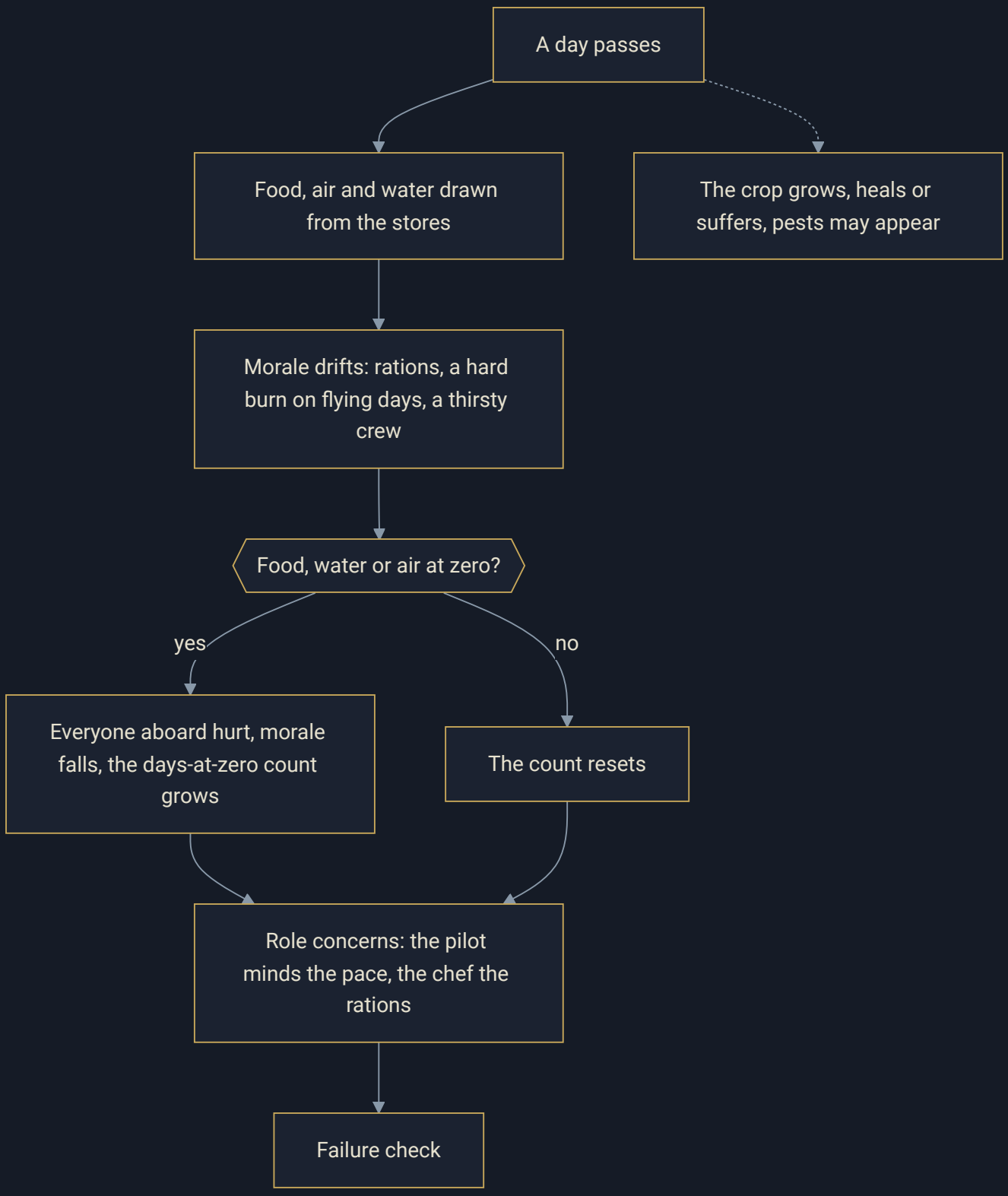
The AI's **callbacks** pick the weightiest memory that's waiting, at most one every other stop, and one is guaranteed after a few quiet legs. At the end, the **Ship's Record** lists up to five of the voyage's most notable choices as epilogue lines, in the order they happened.

Minigames from events

Some choices launch a minigame: a medical emergency, bracing against a solar surge, sealing a hull breach, a scan, an engine sequence, salvage or foraging. The event writes its own reward for each result (excellent, adequate, insufficient), and that is exactly what lands; the minigame's briefing and result card show those rewards, not a free-play haul. A medical emergency plays on a real patient (the worst hurt, the captain included) at their real HP, and the result lands on them. See Minigames and the Stop menu for how quitting works.

The ship's daily clock

Time only moves when something spends it, and every way of spending it goes through the same daily clock: a leg's flight, an outcome's time cost, a Stop-menu job, each day of REST or WAIT, treating the crop, each day of a fabricator print. So the same rules apply whether a day is spent flying or fixing.



One day aboard. The crop's day is worked out the next time anything looks at it, using the same day-by-day rules.

Food, water and air

- **Food** depends on rations (VENDING MACHINE eats half, PIG OUT half again as much), on mouths to feed (the captain, each crew member, any passengers; the engineer eats for two, though a newcomer in the engineer's role eats like anyone else), and is trimmed by a chef aboard and an insulated cargo bay.

- **Air** is a slow draw set by the trail. An O2 recycler halves it, and a healthy crop makes some of the air itself.
- **Water** is one tank that both crop and crew drink from, at the trail's rate. The **water split** in Orders decides how it's shared. A botanist or a water purifier saves some of the crop's share, and the tank feels that saving.

The water split, both sides

The split is one value shown in two places (Orders and the growbay). The *crop's* side feeds the crop model: well under the crop's need slows growth, far under it hurts health, far over it rots the roots. The *crew's* side compares the crew's share with what the crop's need leaves them: the further short, the more morale it costs each day, and at the driest end people also lose a little HP each day. The split can't be pushed all the way to either side.

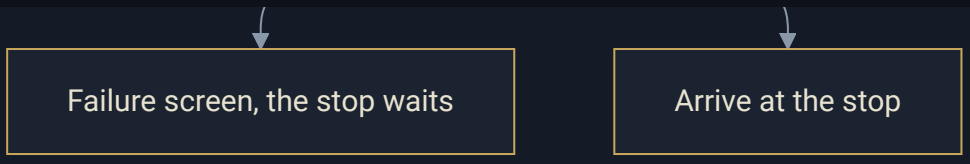
Morale drift

Every crew member's morale moves together with the group changes: rations (short rations sag, a pig-out lifts), a hard burn on flying days (LUDICROUS and PLAID tire people; LIGHT SPEED rests them), and the crew's water share. Once per leg, ARIA adds a little cheer, a chef who has food to cook adds a little more, and a healthy tomato or zinnia crop lifts spirits (sick zinnias drag them down). Then the role concerns add personal reactions on top (see The crew).

A leg

Flying a leg does a few things that a plain day doesn't:


RESUME
Everyone heals a little, more with a medic
Fuel for the leg: the trail's budget per leg x pace x pilot and drive coil
The leg's days pass, a fraction carried to the next leg
At PLAID the drive heats up and may strain
Per-leg perks: ARIA, the chef, solar array, purifier, soil sampler, crop mood
Mishap roll: maybe a popup, never on the hop into the destination
Failure check
a failure
all fine



What one RESUME costs. The final hop into the destination can't fail you for an empty tank: you made it.

Each trail has a travel-day budget and a fuel budget, spread evenly over the legs you'll actually fly (a fork's detour included). Pace scales both: faster means fewer days and more fuel. LIGHT SPEED takes 1.35× the days for 0.6× the fuel; LUDICROUS 0.78× the days for 1.8× the fuel; PLAID 0.55× the days for 4× the fuel (ruling 148: at 3× PLAID ran dry, let the Adrift scene refuel it from the hull plating, and was still the best way to play). PLAID also builds up **drive heat**: each PLAID leg in a row makes a strain more likely (the chance doubles each time), a slower leg cools it a step, and LIGHT SPEED rests it fully. A strain means the next stop draws an engine-strain event.

Mishaps

Every leg also rolls for a **mishap** (ruling 142): something nobody chose, from a list of 21 in `modules/mishaps/mishaps.json`, rolled by `engine/js/mishaps.js`. The base chance is 6% a leg on Lunar, 8% on Mars and 9% on Interstellar, times the pace (×0.7 at LIGHT SPEED up to ×1.7 at PLAID). There's never one on two legs running, and never on the leg into the destination. Each mishap has three tiers; how often each comes depends on the trail (Lunar 72% minor, 26% moderate, 2% severe; Mars 40/42/18; Interstellar 24/44/32), and their costs scale by trail too. A tier can have a *fixed* version that plays instead when the right role is aboard and comes through: a crew specialist always, a captain of that background or a wounded specialist half the time. The engineer covers the hull, the air and the tanks, the pilot fuel, the chef the galley and the stores, the botanist the growbay.

The hull changes the odds (ruling 143). Below 50 hull the chance of a mishap is ×1.5, hull mishaps count three times over in the draw, and severe tiers ×1.5. Below 25 those become ×2.4, six times over, and ×3. The popup must be acknowledged (no ✕, no ESC), the Ship's Log records it in red, and a reload shows the same popup without applying anything twice.

Wear

The hull doesn't wear out on its own; it only loses points to events, fights and the odd drive-coil misfire. The fabricator wears with every print and breaks at full wear (it misfires more as it wears).

When something runs out

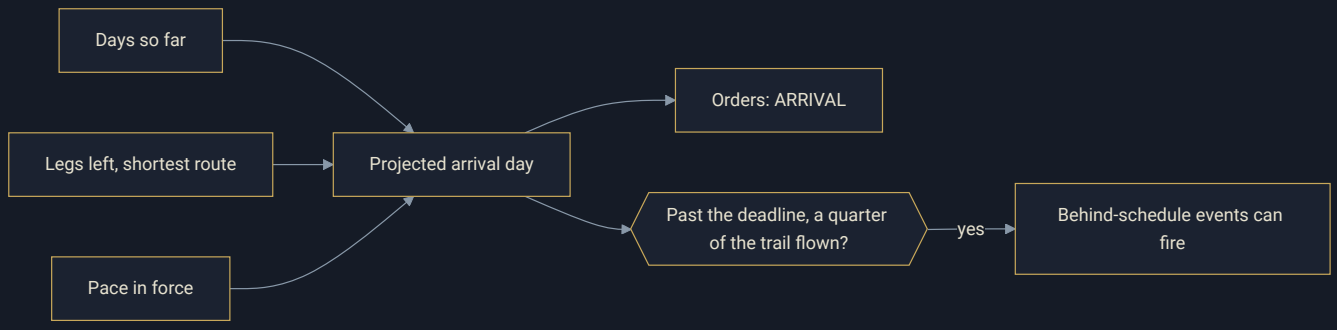
A day at zero food, water or air hurts everyone aboard, the captain included, and drains morale. Each resource keeps a count of days in a row at zero, and enough of them in a row is a failure of its own: starvation, dehydration or suffocation. Air runs out fastest, food slowest. There's a grace rule: the span in which a resource first runs dry can never by itself reach the failure, so you always get at least one stop to react. The count resets the moment the resource is above zero again. (Inside the Adrift scene the same failure is the Adrift ending instead; see Failures.)

The hover boxes use the same arithmetic (ruling 147). A choice whose days would empty a supply says *Water runs out after 4 days* in red, counting what the choice itself takes first. One whose days would reach the failure for a supply that's already gone says *No water aboard: 2 more days without it ends the run*. Thanks to the grace rule, that second warning only ever appears when the supply is already at zero. Stop-menu jobs with a set time and each day step of a wait get the same lines.

The deadline and the behind-schedule warning

Each trail has a deadline, a little longer than a careful run takes. Passing it never ends a voyage: the AI says so once, the ship keeps flying, and the score pays for every day late.

The game keeps **one arrival projection**: days so far, plus the legs still to fly by the shortest route, each at the current pace plus an allowance for a typical stop. Orders shows it as ARRIVAL. Once a quarter of the trail is behind you, if that projection lands past the deadline, the behind-schedule situation turns on and its events can fire. Because both read the same projection, the warning fires exactly when Orders shows a late arrival.



One projection, two readers. Changing pace in Orders changes both at once.

The crew

The ship carries the captain and at most two crew. Each crew member holds a **role** (botanist, engineer, medic, pilot, chef, xenobiologist, diplomat) with a passive skill that is always on and an active skill used from the dossier, which recharges when you dock at a station.

Health

Everyone aboard has HP, the captain included. Below the **wound line** someone is *wounded*: their active skill is offline and their passive works at half strength (a wounded captain's own hands count half on hands-on jobs). Further down they're *too hurt to help* with crew-gated choices. At zero they die. Everyone heals a little each leg, more with a medic aboard and more again for the wounded with a medical scanner. The medic's Treatment, MEDBAY at a station, medkits and healing outcomes do the rest. A medic aboard also softens the direct damage people take, from poisoning, thirst, going without, or a fight.

A **festering wound** (a bad wound left open, ruling 145) loses 7 HP every leg until it's treated: a medic, a heal, the MEDBAY, a medkit or rest that brings the person back over the wound line closes it. That goes for the captain too, so an untreated wound can kill before the next station's crisis ever fires. Some hard and ultra scenes also put the captain personally in harm's way.



A person's health. A death risk can be turned into a bad injury by a medic's training or a medkit.

Specialists and the captain

The captain has a background that matches one of the roles (or merchant, academic, veteran). It brings a starting bonus and that role's passive at **half strength**. A captain and a crew member of the same role don't stack: the better of the two counts. The captain can't stand in for a missing crew member on crew-gated choices, with the xenobiologist exception mentioned earlier.

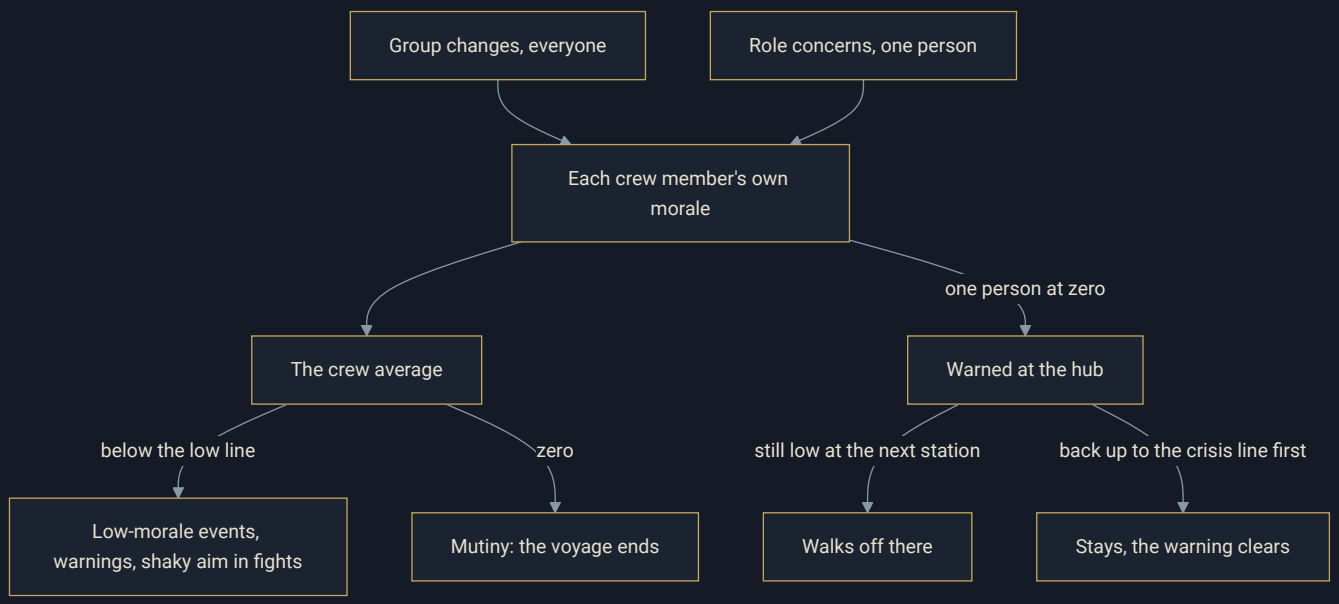
Morale

Morale is **per crew member**; the captain has none. Group changes (outcomes, REST, rations, pace, water, deprivation, the effort tax, fights) hit everyone. A few outcomes name one person: the speaker, a role, or whoever is lowest.

On top of that, each role **minds its own part of the ship**, and only the person in that role feels it:

- the botanist: the crop losing or regaining health, and a harvest;
- the engineer: hull damage, and a finished repair;
- the medic: anyone hurt, and anyone lost;
- the pilot: days at a crawl (and enjoys a fast burn);
- the chef: short rations, and an empty galley;
- the diplomat: any fight at all;
- the xenobiologist: alien finds;
- everyone: being wounded themselves.

This is what lets one person drift away from the group. A lone crew member's concerns count half, so the whole crew's mood doesn't swing on one person's worries. The crew strip's MO bar tooltip says what last moved them.



Morale, from the individual up. A small passive lift isn't enough to cancel a warning; it takes REST, a good outcome or better rations.

Effort

Crew are also hands. On an effort check, a listed specialist contributes their full skill (half if wounded), everyone else adds general help, and the captain's own hands always count. That's why a two-person crew with the right roles

finishes jobs a solo captain struggles with.

Deaths and leaving

Whatever road a crew member leaves by (death, the airlock, a dismissal, desertion, stepping down for a stowaway), they take everything that was theirs: the role, HP, skill charges, an open wound, morale, and any skill they had armed but not spent. A death is written into the Ship's Log as a memorial and sets a lasting flag. The voyage goes on as long as the captain lives. If the captain dies, it ends: *captain lost* with crew still aboard, or the autopilot ending if there is no crew (which names any passengers still riding, rather than calling the ship empty).

Stowaways

About a third of voyages launch with a hidden stowaway, one of seven people, each with their own story and (for most) a specialty. Their **reveal window** opens once about a fifth of the route is flown; from then on the events that find them get a stowaway pull, and some ordinary events have a stowaway variant that waits for the same window. Found, they can be welcomed, refused, or (for one with a specialty, while you have crew) welcomed by having a crew member step down: the one in the stowaway's own role if there is one, otherwise someone at random. Welcomed, they join as crew in their own role if a place is open and nobody holds that role; otherwise they ride as a passenger to the end of the voyage. Someone who stepped down rides as a passenger to the next berth. A welcomed stowaway's follow-ups (a scene about them settling in, the AI's callbacks about their work) wait on them being aboard, and stop if they die or leave.

Rescues

Three events can bring a named person aboard: Mira Kastel, the engineer in *A Pulse*; Hester Quill, the botanist behind the one pod shut from the inside in *The Pod Garden*; and Tal Okafor, the off-shift dock engineer at *The Long Dark Bar*, once you've bought them a drink. Each joins the crew if a place is open and their role is free, and otherwise rides as a passenger to the next berth, where their thanks (gold, and a little morale for the crew) is paid. These events carry a rescue pull, stronger once the crew is short-handed (see Pulls). Once Tal is aboard, the bar's hire is spent and stops pulling.

Passengers

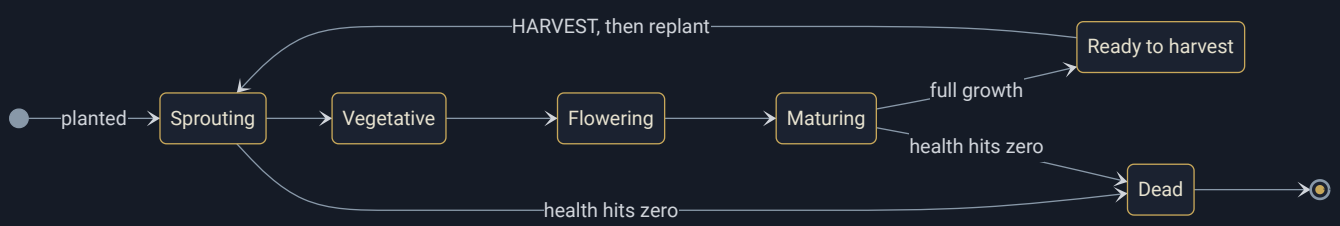
A passenger is anyone aboard without a crew place: a rescued rider, a crew member who stepped down, or a welcomed stowaway with no place. They don't work, and they don't count as crew, but they eat like crew, so the food draw, the Orders projection and the hold's food line all count them. The cruise hub names them in a passenger line at the bottom left of the scene, so the crew strip never shifts. Riders leave at the next berth; the stowaway stays. With no crew left but a passenger aboard, the captain isn't alone: the pause terminal says passengers are aboard, and every arrival ending has its own crew paragraph that names them.

Newcomers' faces

Every stowaway and every rescued crew member has their own portrait sheet: their face on the crew strip, the dossier, crew talking-head scenes and the Ship's Log, and the stowaway's face in the scene that finds them. The sheets are still images for now, so the face holds still while a line types out. In the growbay a newcomer tending the crop stands as their portrait, without the roster's tending animation.

The crop

The crop is the mission. It lives in the growbay (not in cargo) and has two numbers: **growth** and **health**.



Growth stages. Death can come at any stage; only two are drawn here to keep it readable.

Growth and health, day by day

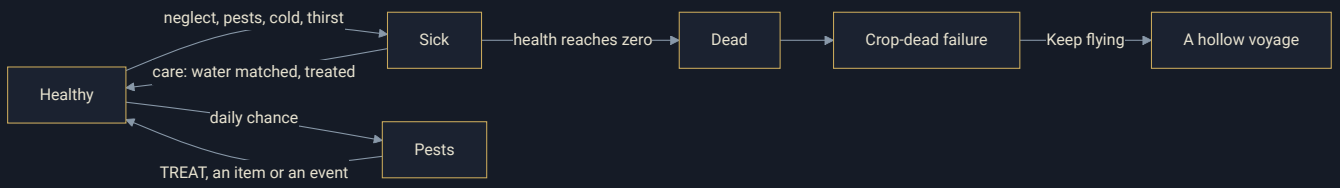
Each day the crop grows toward maturity at a rate set by its own maturity time, multiplied by its health and by every active modifier. Health moves by the sum of the modifiers' health effects. Every modifier the growbay lists is real and carries the numbers the day uses:

- **Water**: matched to the crop's need it slowly heals; under it growth stalls and, badly under, health falls; far over it rots the roots; a dry tank hurts it most.
- **Temperature**: the stop's environment (the void runs cold, anomalies hot) against the crop's ideal band.
- **Light**: nebulae and the void are dim, which slows light-hungry crops.
- **Tending**: whoever tends the bay adds growth (a botanist most, and a bonus to the yield).
- **Talking to it**: after a station botanist's advice, a small lasting growth bonus.
- **Pests**: slow growth and steadily eat health until treated.

Each crop has its own weaknesses and strengths that scale these (fragile zinnias, water-sensitive tomatoes, slow-recovering sweet potatoes, wheat's vulnerability to blight, soybeans that recover faster after pests). Tomatoes are also drama-prone: now and then a leg brings a small growbay incident that nudges their health or the crew's mood either way.

Sickness, pests and treatment

Below its sick line, a crop is *sick*: the AI says so, crop-sickness events join the pool, and the botanist frets. Pests arrive by a small daily chance once the plant is established. They can be cleared by TREAT in the growbay (a day of work, which a botanist spares you), a crop treatment from the hold, or an event outcome. The botanist's Greenhouse Protocol restores some health directly, and a botanist makes crop recovery from events go further.



The crop's health. A healthy crop also helps the air supply and, for tomatoes and zinnias, morale.

Harvest

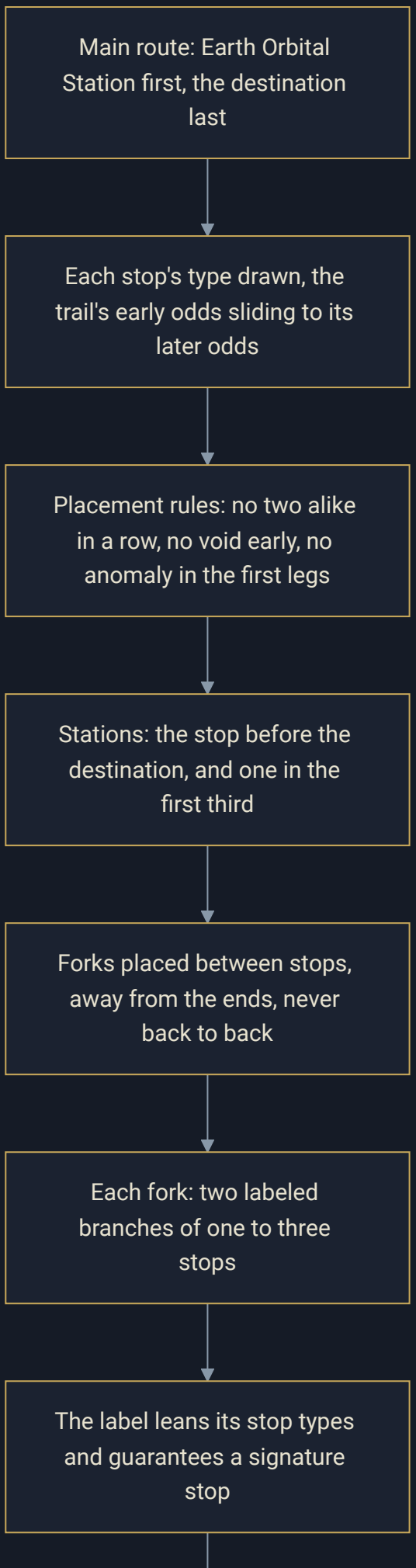
At full growth the crop is ready. Harvest (from the growbay or the crop card) pays its yield: food for most crops (anything past a full store is packed into ration bricks, if there's room), morale for zinnias. The bay replants at once and starts a new cycle.

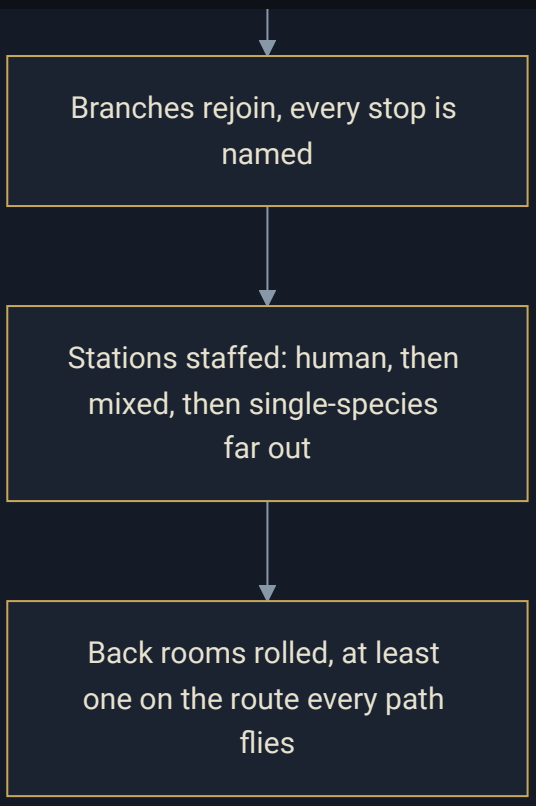
The crop dying

If health reaches zero, the crop-dead failure appears. It offers **Keep flying**. Accepting turns on a lasting "crop dead" situation, so a few aftermath events can appear, and the arrival will be *hollow* no matter how well the rest went.

The map

Each trail has a fixed length, a number of forks, and tables saying which kinds of stops are likely early on and later. At launch the route is generated fresh.


Main route: Earth Orbital Station first, the destination last
Each stop's type drawn, the trail's early odds sliding to its later odds
Placement rules: no two alike in a row, no void early, no anomaly in the first legs
Stations: the stop before the destination, and one in the first third
Forks placed between stops, away from the ends, never back to back
Each fork: two labeled branches of one to three stops
The label leans its stop types and guarantees a signature stop

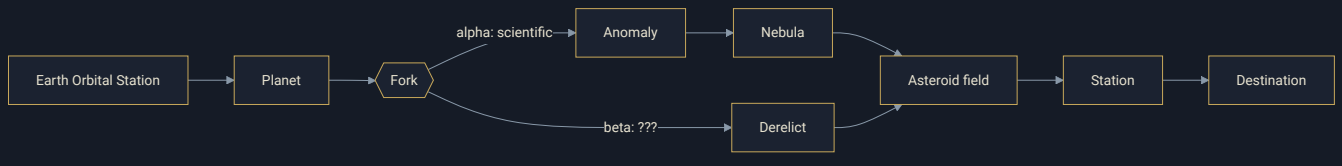


How a route is built. Every rule is about keeping the trip readable and fair; the dice do the rest.

Forks and branch labels

Each fork has two branches, and each branch gets a label: hostile, lucrative, safe, corporate, derelict-heavy, scientific or agricultural. About a third of forks hide one branch's label ("???"); a hidden label still shapes its branch. Labels are real in two ways:

- They **lean the branch's stop types** (a derelict-heavy lane fills with derelicts, a corporate one with stations, an agricultural one with planets), and every labeled branch is guaranteed at least one of its signature stops, so a label never lies.
- They **lean the events** on every stop of the branch: hostile toward fights and hard scenes, lucrative toward rewards, safe toward easy and peaceful ones, scientific toward science, agricultural toward crop and food, corporate toward stations and traders.



A small example route with one fork. Real trails are longer and the Interstellar trail has several forks.

Species space

Each run crosses one alien species' space: the **voyage species**, one of six (insect, reptile, robot, rock, water and the unknown ones), rolled at launch, named by the AI at departure and saved with the run. The route has zones:

- The first stretch is **human**: its stations are human, and alien species' events don't fire there.
- Then **mixed**: stations where anyone works.
- On Mars and Interstellar, the last third is **far out**: a far station stays mixed or belongs to one species. The last station before the destination always belongs to the voyage species, because every path passes it. Each other species holds at most one station a run, the voyage species at most two. Lunar's route never reaches the far zone, so it has no single-species stations.

On the map a single-species station is labelled with its species in place of "Station" ("Reptile 11"), and its description says whose it is. Its crew and traders are all that species (which suspends the usual rule that no two people at one stop share a species); the bartender is the one exception, the same kind at every station. Everyone else you meet is rolled by distance: mostly human near home, mostly alien far out, with a lean toward the voyage species. Species events follow the same map. They fire only at their own station or a mixed one (open space past the human stretch is fine too), the voyage species' draw three times as often, and each species has one scene written for a station wholly its own, drawn six times as often there, so it is usually what plays when you reach that species' station.

Each species also has a **standing item**: a token some of its events hand you (a Hive Lozenge, a Basking Chit, an Extended Warranty and so on; the human one is a Stamped Exemption). It takes no cargo slot and unlocks choices in that species' other events while you carry it.

Back rooms

When the map is built, each station past the human stretch (the destination aside) keeps a back room about half the time, and every run has at least one on the route every path flies, never only on a detour. The map doesn't say which. See The black market.

Scanning ahead

SCAN draws the events waiting at the stops ahead *now* and locks them in, so what the scan reports is what you meet. A good scan reads the next stop (every branch of a fork); a perfect one reads two stops deep and also turns on outcome previews for every choice at those stops; a poor one reads only static. The report names each event and how hard it is. A locked-in event is checked again on arrival: if it can no longer happen (you sold the thing it needed, say), the stop draws afresh. The ship also announces some stop types as you approach (good mining ahead, a derelict, an unidentified signal).

Minigames and the Stop menu

The Stop menu offers **verbs** that depend on the kind of stop: MINE in asteroid fields, SALVAGE at wrecks, FORAGE (or GATHER, HUNT, SCAVENGE) on planets and rocks, CONDENSE in nebulae with the right upgrade, SCAN, REPAIR, REST, the FABRICATOR, the station services, and BACK ROOM at a station that keeps one. TRADE appears when a trader is in range (every station has one; elsewhere each stop rolls once whether one is around, likelier near civilization); otherwise WAIT FOR TRADER takes its place. A verb with nothing to work on, like REPAIR on an intact hull, isn't offered at all.

How a job works

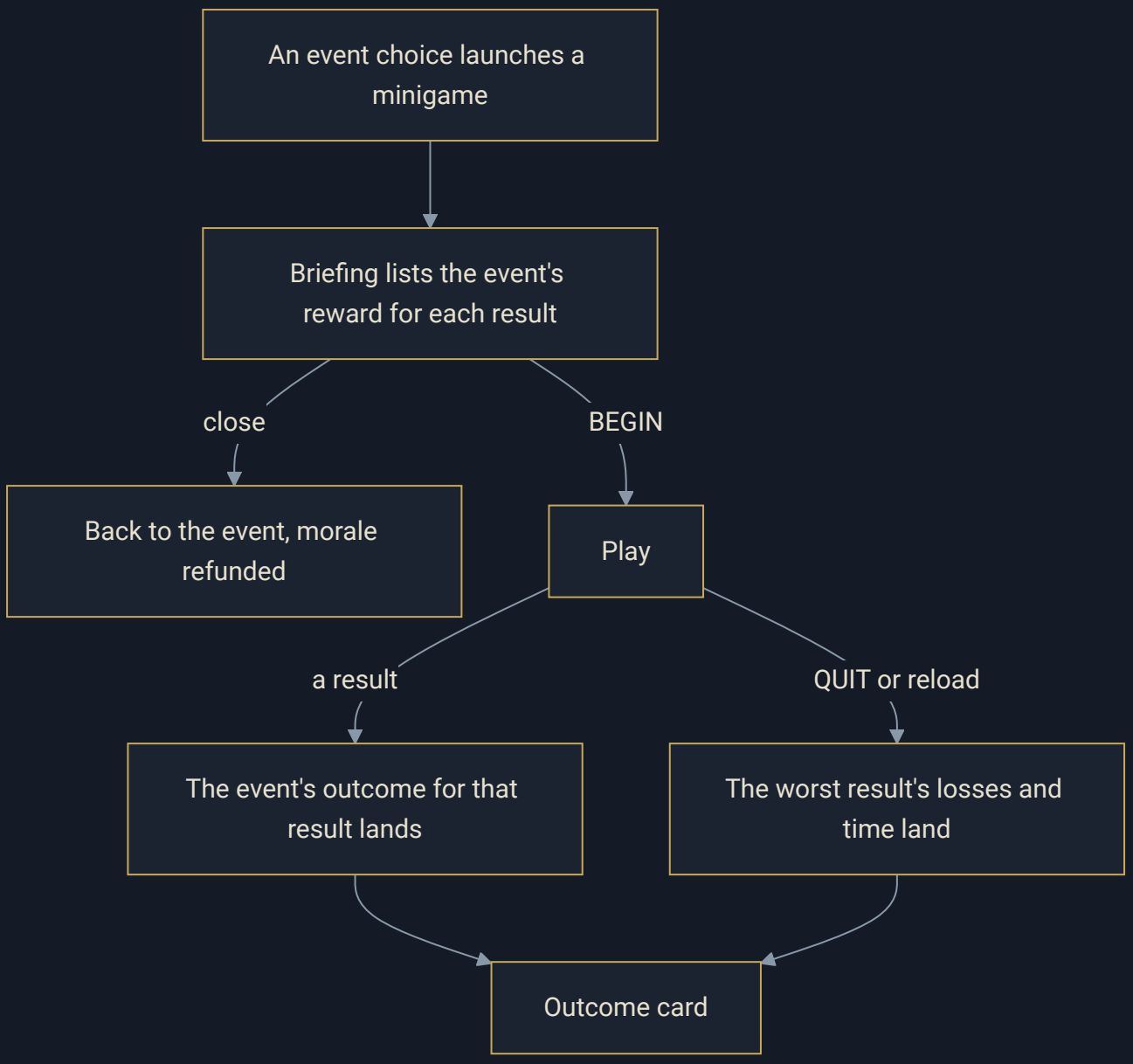


A Stop-menu job. Backing out before BEGIN costs nothing; after BEGIN, the only way out is through the time it costs.

- **The effort tax** is taken when you click, but held: closing the briefing (or leaving the mining field before your first run, or walking out of the engine room untouched) hands it back.
- **BEGIN** is the point of no return. The game marks the save, so a reload from here is a quit.
- **Quitting** keeps nothing, not even what was already in hand, and still spends the time. Harm already taken (a poisoning while foraging) stays taken.
- **The result card** grades the run (excellent, adequate or insufficient) and lists what lands, as it will really land: a full food store or hold is shown as left behind.

Most jobs take a day. MINE takes a day per run. REST and WAIT pick a number of days and tick through them one at a time, with the full daily clock each day; waiting outside a station isn't safe: each day can bring a pirate, a distress call or an anomaly, which ends the wait and opens that event (a broken rest keeps the morale for the days it got). Station services (MEDBAY, DRYDOCK, FAB SERVICE) cost gold, not days, and the back room takes no days either. The fabricator charges per print and runs its days only while you watch it; closing it holds the job.

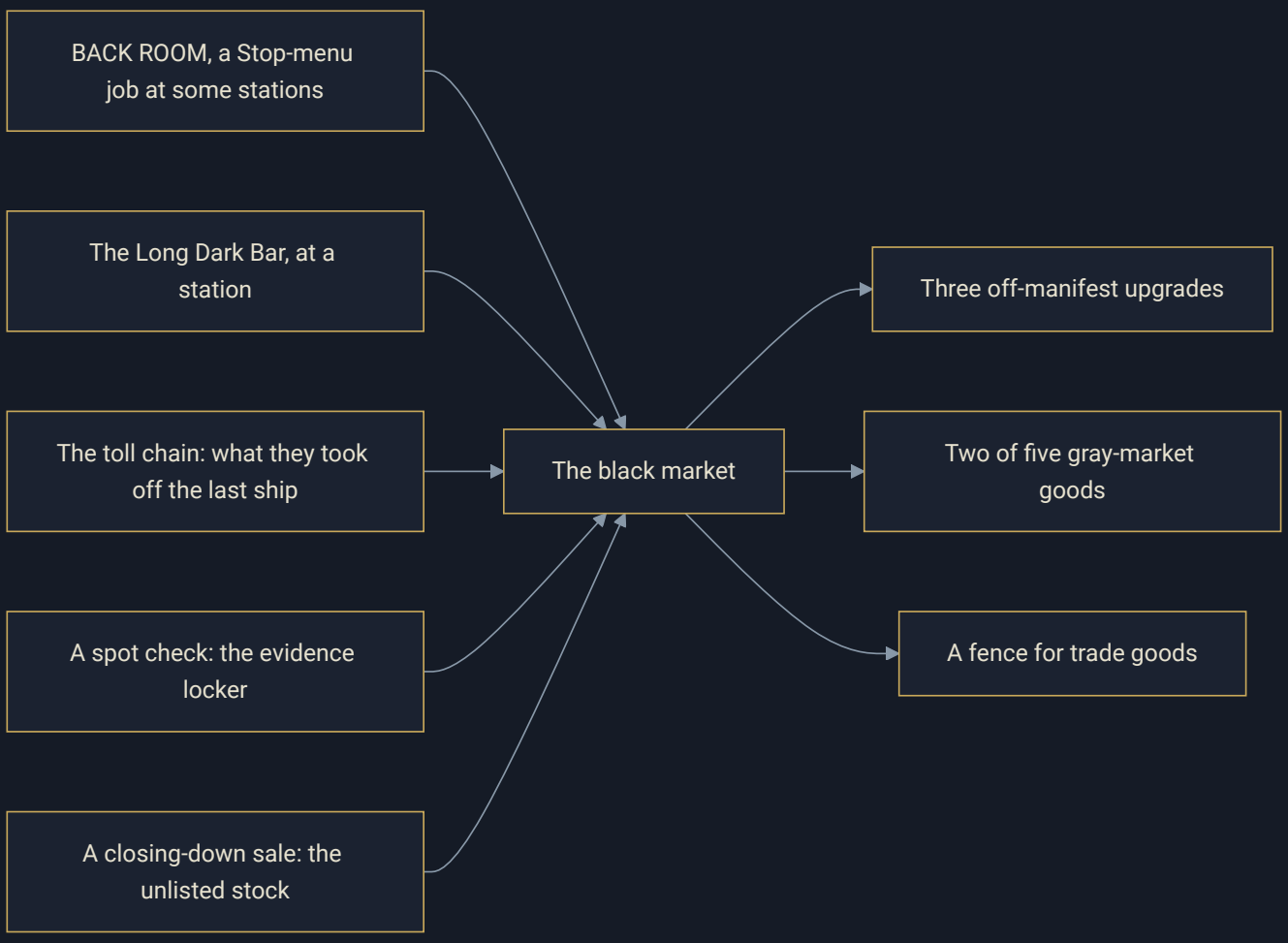
Minigames launched from events



Event minigames differ in one way that matters: quitting can't beat playing badly, because a quit takes the poor result's losses.

The black market

The black market is one screen with several doors. It sells what no honest shop shelves and buys what an honest trader would rather not ask about.



The doors. The three lane encounters each have their own greeting and farewell, and their own pair of gray goods.

- **The back room** is a Stop-menu job at a station that keeps one: two effort pips, no days, no event. It opens over the hub, which dims behind it. It is shut, with the other station services, at a station you waved off or flew past.
- **The shelves.** Three upgrades no honest shop sells (a drive coil, an O2 recycler, a field repair fabricator), minus any you already carry, and two gray-market goods.
- **Who sells decides the pair.** Each lane encounter shelves its own (the toll chain's stolen fuel and foam, the inspector's seized medkits and rations, the closing-down trader's oxygen and foam). A single-species station's back room sells that species' pair, with its own greeting. A mixed station's back room and the bar keep a random pair, the same all stop.
- **The fence** buys trade goods only, at half again what an honest trader pays. It doesn't take materials or ordinary gear.
- **Discounts.** CHIP's and a Merchant captain's discounts apply here as at any market.

Gray-market goods

Each is a bigger refill than its honest counterpart, used from the hold, and sometimes it backfires. The tooltip names the harm a backfire does, but never the odds. They are priced so that, on average, the refill costs about what the honest item does. They stack, so you can buy several at once.

GOOD	WHEN IT WORKS	WHEN IT BACKFIRES
Bulk O2 Tank	O2 +35	+8
Hot Fuel Cell	FUEL +35	+10, and the hull takes 6
Hull Foam	HULL +30	+10
Unlabeled Ration Crate	FOOD +25	+5, and the crew loses 3 morale
Gray-Market Medkit	The worst hurt aboard healed to full	Half the healing, and the crew loses 3 morale

A backfire's harm is real harm: if it takes the hull or the crew's morale to zero, the voyage ends there, as it would anywhere else.

Combat

A fight starts from an event choice that opens fire, or from a ram that doesn't get through (ramming works most of the time; when it doesn't, you take a knock and the fight starts anyway). Fleeing resolves straight to its own outcome. Every fight is fought from the bridge: their ship in the window, their captain on the feed. The fight tracks its own hull and crew-morale numbers, then hands the net result back to the ship.



A ship-to-ship fight. Free actions (hull patch, boost shields) don't end your turn. A tribute that's taken ends the fight on the spot.

Your options

- **Fire**: click their ship to aim. A shaky crew can miss; a keen one lands more critical hits; a xenobiologist who identifies the enemy adds damage.
- **Evasive maneuvers**: soften the next hit (less so with a frightened crew); a pilot adds a chance to dodge it entirely.
- **Hull patch** (engineer, once per fight, below half hull) and **boost shields** (with a shield module, once per fight, blocks the next hit) are free actions.
- **Offer a tribute** on the opening turn: the botanist's harvest or the chef's cookies, each a gamble costing food either way. Pirates who took the cookies remember them, and let you pass if you meet them again before the next station.
- **Attempt escape**: the drive spools for a few turns (fewer with a pilot) while they keep firing, though the spool makes you harder to hit. It costs fuel when it jumps.

Their side

Pirate ships vary by species: some are tougher, and some never surrender. On longer trails they hit harder and take more punishment. The diplomat softens their opening shot. A badly damaged pirate of a surrendering species usually

offers terms, or sometimes just flees. If the crew's morale collapses mid-fight, they stop firing, and the only order left is to stand down. With REX aboard, the first hit of every fight lands on point-defense instead of the hull.

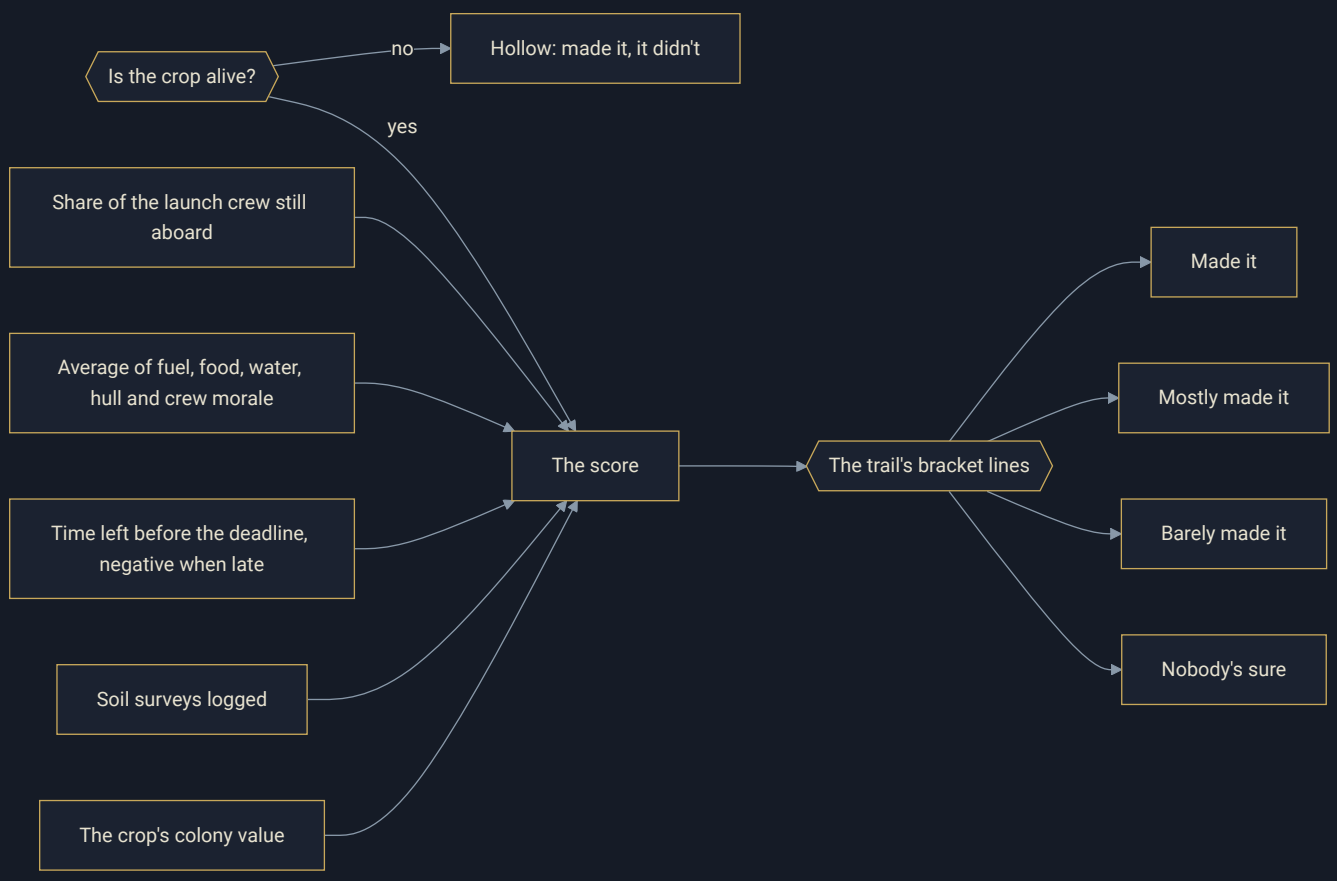
How fights end

- **Destroyed**: salvage the wreck (full loot and a bonus roll, more days) or leave it (partial loot, back sooner).
- **Surrender accepted**: part of the loot, a morale lift. **Finished them**: the full loot, a morale hit, and a flag the galaxy remembers.
- **Tribute taken**, **escaped**, or **they fled**: no loot, the fight's costs only.
- **Stood down**: they take a toll of fuel and gold (a species that won't accept a yield treats it as a defeat).
- **Defeated**: the ship limps away with hull barely holding, a share of the cargo is taken, and people may be hurt or, rarely, killed (a medic and high morale make that rarer).

Whatever happened, the diplomat minds the fight, the engineer minds each hull hit, and the few stops afterward count as a post-combat window for aftermath events.

How a run is scored and ended

Reaching the destination ends the voyage with an **arrival**. Its tier comes from a score worked out at that moment, from the crew who made it, what's left aboard, and the time.

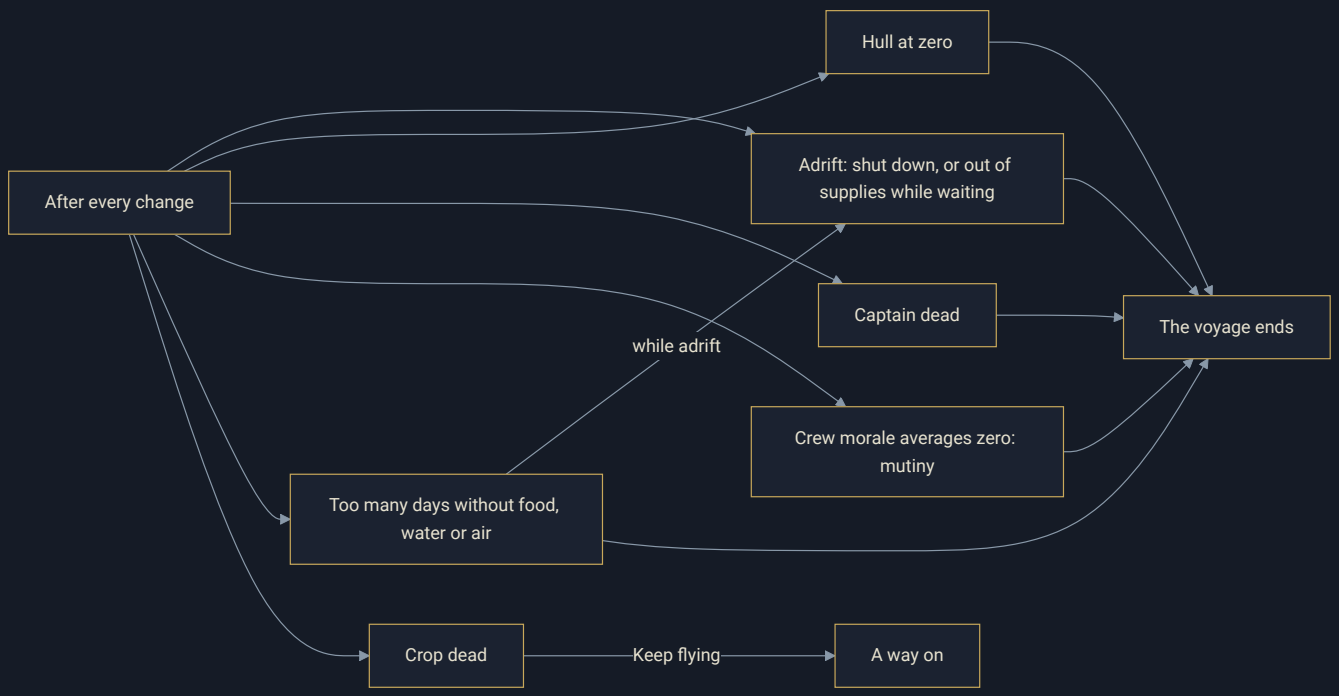


The arrival score. Crew and supplies make the base; time is a big swing either way. Air isn't part of the average.

- **The crop** is the gate: a dead crop makes the arrival hollow whatever the numbers say. A living crop multiplies the score by its colony value (tomatoes only count if they arrive in decent health); living zinnias add nothing to the score but change the ending's final paragraph. The crop's condition (thriving or sick) also changes the ending's text.
- **Crew** is the share of the crew you launched with who are still aboard (a captain who chose to fly solo gets half credit). Newcomers count, but the share can't go past full, and passengers aren't crew.
- **Supplies** are the average of what's left, crew morale included.
- **Time** is the share of the deadline left in hand. It is weighted heavily, and the same rate runs backward once you're late, so every late day costs as much as an early day earns. Late arrivals also get endings that don't claim you were on time.

Each trail draws its own bracket lines, calibrated so careful runs spread across the tiers by how well they went on *that* trail. The ending text is then assembled from the tier, the crop's condition, how the crew fared (a full crew, a partial one, the captain alone, or the captain with only passengers left, whom it names), the destination, and the Ship's Record of your notable choices. There are twelve arrival endings: two for each of the four tiers, two that read across every tier, and two hollow ones.

Failures



The ways to lose. Only a dead crop offers a way on; an empty tank opens the Adrift scene first. They come to nine failure screens: food, water and air each have their own, and a dead captain reads as *captain lost* with crew aboard or *autopilot* without.

Adrift (ruling 144): an empty tank is not a failure by itself. The leg that runs the tank dry still arrives, so a station's fuel or a trader can still save the ship. Trying to fly on with nothing in the tank opens the Adrift scene (`event_adrift_001` in `events_crisis.json` , never drawn from the pool): burn hull plating (−25 hull, +15 fuel), crack the water (4 days, −35% water, +12 fuel), trigger the emergency beacon if carried (someone always answers: a friendly hauler with free fuel, a corporate tug that bills 40% of your gold, or people who take half of it at gunpoint), call for help and wait (40% a tug tows you clear for 30% of your gold in 6 days, 20% the wrong people in 5 days, 40% nobody in 10 days), or shut it all down. Shutting down ends the run as the Adrift failure, and so does running out of food, water or air while waiting in the scene: no starvation, thirst or suffocation screen, but Adrift with its own text, and no beacon on offer, because nobody is left to rescue. **Keep flying** after a dead crop carries on toward a hollow arrival, and the stop the failure interrupted still plays.

Records and bests

A voyage goes into the records once its ending is final: at once for an arrival or a terminal failure, or, for a failure that offers a way on, when you accept it (by reading the report, or by launching a new voyage over its save). The records keep the last ten voyages, newest first, and each destination's **best** arrival. Only arrivals with a living crop count for a best, a higher score wins, and on a tie the faster voyage does. Beating the best shows NEW BEST on the report. A best's crew reads "crew 2/2", or "crew 2 (launched with 1)" when newcomers outnumber the losses. Records live apart from saves, so starting over never clears them. They also feed the game: a voyage that ended in failure can turn up in a later run as a wreck.

The AI companions

The AI you pick at setup is the voice of the whole voyage. Almost every line the ship says passes through it.



Where the AI's voice turns up. At a stop, its lines also show in a bubble over the ship: the newest three, with a first-run tip outlasting the chatter. RESUME clears the bubble; the lines stay in the log.

- **Events** carry a line written for each AI, so the same scene reads five ways.
- **The Ship's Log** records choices, outcomes, arrivals, fights and deaths in the AI's own register.

- **Tips** come one per new stop, the most urgent first (pests, a wound, low supplies, low morale), then an introduction to one system at a time. Each topic is said once per voyage, and the pause menu can switch them off.
- **Warnings** fire when a supply or the crew's morale crosses its low line going down, and again at a critical line. Recovering and dipping again warns again.
- **The deadline** passing is announced once, in the AI's voice.
- **The voyage species** is named at departure, in each AI's own words.

What each AI's perk does

AI	PERK, IN PRACTICE
ARIA	A little morale for every crew member on every leg. Small, steady, relentlessly upbeat.
MARV	Shows the costs of the costliest choice in every encounter before you pick.
REX	The first enemy hit of every fight is blocked, and the arcade minigames start with an extra life: one more mistake forgiven in the engine sequence, the first surge absorbed when bracing, more time on a breach, a slower drain on the salvage tractor.
CHIP	Cheaper stations: market wares (the black market's too), MEDBAY, DRYDOCK, FAB SERVICE, parking fees and gold spent inside station events.
AJOY	Hindsight: after an outcome, names a choice you didn't take and what it would have done.

Perks never change which events appear (though a few events are written for one AI in particular, and only turn up with it aboard). They change what you know (MARV, AJOY), what things cost (CHIP), or how hard the ship is to hurt and how cheerful it is (REX, ARIA).

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